Solution of the Reynolds-averaged Navier–Stokes equations for transonic aerofoil flows

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SUMMARY

A computational method to predict the viscous transonic flow development around two-dimensional aerofoil sections is described. The Reynolds-averaged Navier–Stokes equations applicable to turbulent flow are discretised in space using a cell-centred finite-volume formulation. A multi-stage, explicit, time-marching scheme is used to advance the unsteady flow equations in time to a steady-state solution. Turbulence closure is achieved using either the Baldwin–Lomax algebraic model, or a one-equation model based on the turbulent kinetic energy equation. This latter equation is solved using essentially identical procedures to those for the mean-flow equations. Results are presented for the RAE 2822, RAE 5225, CAST 7 and MBB-A3 transonic aerofoil sections. The relative performance and limitations of the two turbulence models are discussed.

1. NOTATION

C_D, $C_D(T)$	total drag coefficient
$C_D(P)$	pressure drag coefficient
$C_D(V)$	viscous drag coefficient
C_L	lift coefficient
$C_m(\frac{1}{4})$	pitching moment coefficient (about ¼ chord)
c	aerofoil chord
c_{fe}	skin friction coefficient
$\mathbf{D}$	vector of artificial dissipative terms, Equation (28)
E	total energy per unit mass
F	function in Baldwin–Lomax model, Equation (51)
$\mathbf{F}^i$, $\mathbf{G}^i$	convective flux vectors, Equation (4)
$\mathbf{F}^v$, $\mathbf{G}^v$	viscous diffusive flux vectors, Equation (7)
H	total enthalpy per unit mass
$\mathbf{H}$	flux tensor, Equation (3)
h	cell volume (area)
$\mathbf{i}$, $\mathbf{j}$	cartesian unit vectors
k	turbulent kinetic energy per unit mass
M	freestream Mach number
P	pressure
P_k	k transport equation production term, Equation (59)
Pr	laminar Prandtl number
Pr_t	turbulent Prandtl number
$\mathbf{Q}$	inviscid flux integral, Equation (19)
q_x, q_y	components of heat-flux vector, Equation (11)
R	freestream Reynolds number
$\mathbf{R}$	vector of residuals, Equation (35)
S_k	k transport equation source term, Equation (58)
$\mathbf{S}^v$	vector of source terms, Equation (8)
s_{xx}, s_{yy}, s_{xy}	components of mean-strain tensor, Equation (10)
T	temperature
t	time
U, V	cartesian mean-velocity components
U_e	value of U_T at boundary-layer edge
U_T	mean velocity tangential to surface
$\mathbf{V}$	viscous flux integral, Equation (22)
$\mathbf{W}$	vector of dependent variables, Equation (2)
X, Y	cartesian coordinates
$y_{\max}$	surface normal position of $F_{\max}$
y_n	surface normal distance
y^+	surface normal distance, Equation (47)
α	angle of incidence
γ	ratio of specific heats
Δt	time step, Equation (43)
Δt^*	time step for CFL of unity, Equation (42)
ΔX, ΔY	cartesian lengths of cell edge
δ	boundary layer thickness
δ^*	displacement thickness
ϵ	dissipation rate of k
θ	momentum thickness
μ	molecular viscosity
μ_t	eddy viscosity
ν	shock-wave sensor, Equation (33)
ξ, η	local curvilinear coordinates
ρ	density
σ_{xx}, σ_{yy}, σ_{xy}	components of viscous stress tensor, Equation (9)

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Subscripts

a, b, c, d	cell vertices
corr	corrected value
i	inner
i, j	cell-centre value
k	cell-edge value
max	maximum value
min	minimum value
o	outer
t.e.	trailing edge
w	surface conditions
∞	freestream conditions

Superscripts

*	dimensional quantity
ξ, η	curvilinear components

2. INTRODUCTION

Accurate prediction of the viscous flow development around two-dimensional aerofoils at transonic conditions remains an important goal in computational aerodynamics. This is particularly true for modern supercritical aerofoils, involving significant rear loading, for which viscous effects are generally more important than on more traditional aerofoil sections(1). Supercritical aerofoils are designed to feature a weak upper surface shock wave at the conditions of practical interest, so as to minimise wave drag. However, further downstream there results a steep adverse pressure gradient, leading to a thick turbulent boundary layer which is close to separation in the trailing-edge region. Viscous effects can reduce significantly the lift generated by such aerofoils, relative to values predicted using inviscid flow calculation methods.

The use of viscous methods in transonic aerofoil design has been particularly successful for civil applications. This is due to the great importance attached to minimising drag at the design cruise condition. Thus, only weak shock waves can be tolerated, and flow separation is to be avoided. Also, at the high Reynolds numbers of practical interest, the effects of viscosity are confined to relatively thin boundary layer and wake regions. Under such fully-attached flow conditions, with weak viscous/inviscid interactions, computational methods which couple a boundary-layer solver to an inviscid-flow solver have proved to be highly successful, leading to methods such as VGK(2) in the UK and GRUMFOIL(3) in the USA.

VGK and GRUMFOIL solve the full-potential equation for the inviscid flow, coupled to integral solution techniques for the boundary layers and wakes. The full-potential equation assumes isentropic flow and so cannot predict the correct pressure jump across shock waves for inviscid flow, although Lock(4) describes a technique which can compensate for this to some extent. However, it is generally accepted that such methods are restricted to flows involving weak shock waves, for which isentropic flow is a reasonable approximation. More recently, solutions of the Euler equations for compressible inviscid flows have been developed. The Euler equations give the correct Rankine–Hugoniot pressure, density and velocity jump conditions, and so result in entropy production across shock waves. Whitfield *et al*(5), Drela and Giles(6) and Chen *et al*(7) all describe methods combining solutions of the Euler equations with integral boundary layer procedures.

Viscous/inviscid matching methods, as described above, appear to work quite well for fully-attached flows. However, in practice, they may require elaborate smoothing techniques, particularly in the trailing-edge region, to enable converged solutions to be obtained. Further, direct boundary-layer methods break down in the presence of flow separation, and inverse solution techniques are then required; *see* Lock and Williams(8). Classical boundary layer assumptions become less valid for flows approaching and beyond separation, where the effects of surface-normal pressure gradients and Reynolds normal stress terms can be significant. East(9) discusses higher-order boundary layer approximations to the Reynolds-averaged Navier–Stokes equations for viscous flows. Such approximations, in turn, require higher-order matching of the boundary-layer equations to the outer inviscid flow. Nevertheless, it is generally acknowledged that even BVGK(10), the latest incarnation of the VGK method which incorporates such higher-order techniques, is restricted to flows with only rather limited regions of separation.

Nowadays, the increasing power of modern supercomputers, with vector and possibly parallel architecture, enables the nominally-exact governing flow equations to be solved efficiently. However, even the Reynolds-averaged Navier–Stokes equations require a turbulence model to achieve a closed set of equations for the mean-flow development. The accuracy of the resulting predictions is critically dependent upon the performance of the turbulence model. Holst(11), in a review of the AIAA Viscous Transonic Airfoil Workshop indicates that increasing sophistication of turbulence model does not automatically lead to improved predictions, at least up to the two-equation model level. Indeed, the most consistently successful turbulence model at the workshop is that of Johnson and King(12), which is essentially an algebraic model (there are some reservations about this model, however, since it appears to be 'over-calibrated' to non-equilibrium flow conditions, as can be seen in the RAE 2822 aerofoil case 6 and 9 calculations of Coakley(13)). The AIAA workshop also highlights the need to compare the relative performance of different turbulence models within the same numerical scheme, in order to remove uncertainties regarding differences in flow algorithms, computational grids etc.

With the above as background, the present work is concerned with the development of a flexible flow solver for the Reynolds-averaged Navier–Stokes equations. A relatively simple solution algorithm is adopted, so that effort can be concentrated on the implementation and evaluation of a range of turbulence models. Thus, the governing flow equations are discretised in space using the cell-centred, finite-volume approach of Jameson *et al*(14), adapted for application to viscous flows. The resulting semi-discrete equations are solved by an explicit multi-stage scheme, marching in time to a steady-state solution.

The paper describes the implementation of two eddy-viscosity based turbulence models, although the eventual intention is to evaluate models up to the Reynolds-stress transport equation level. It is acknowledged at the outset that the two turbulence models considered are not expected to perform well for flows involving significant regions of shock-induced or trailing-edge separation. Therefore, the current work should be viewed as a validation exercise for the basic flow solver, together with a preliminary evaluation of two relatively-simple turbulence models.

The governing flow equations are presented in Section 3, and their spatial discretisation is described in Section 4. The numerical dissipative terms required to stabilise the solution procedure are discussed in Section 5, with the time-marching scheme itself being presented in Section 6. Two turbulence models and their numerical implementation are discussed in Section 7. Results are presented in Section 8 of viscous flow computations for the RAE 2822, RAE 5225, CAST 7 and MBB-A3 transonic aerofoils. These are all supercritical sections, and so are representative of current practice for transport aircraft applications. The paper closes with some conclusions from the present work in Section 9.

3. GOVERNING FLOW EQUATIONS

The calculation method is based on a solution of the Reynolds-averaged Navier–Stokes equations, written in terms of mass-weighted average variables[21] for application to compressible turbulent flows. The time-averaging procedure used to obtain the set of mean-flow equations introduces new unknowns, the Reynolds stresses, and these are modelled via an isotropic eddy viscosity. Two turbulence models are to be considered, the algebraic model of Baldwin and Lomax[15] and a one-equation model. This latter model involves the solution of an additional transport equation, for the turbulent kinetic energy. In anticipation of the turbulence-model implementation techniques described in Section 7 below, this additional model transport equation is integrated directly into the set of mean-flow equations. Finally, all equations are written in terms of non-dimensional variables, using the scalings given in Equation (16).

The governing flow equations are written in integral form, to facilitate the finite-volume spatial discretisation adopted in the calculation method. Further, the time-dependent equations are to be solved, by marching in time to a steady-state solution. The resulting equations can be written as follows

$$\frac{\partial}{\partial t}\int_{\Omega} \mathbf{W}\, d\Omega + \int_{\Omega_s} \mathbf{H} \cdot \mathbf{n}\, d\Omega_s + \int_{\Omega} \mathbf{S}^v\, d\Omega = 0 \qquad \ldots (1)$$

Ω is any two-dimensional flow domain, Ω_s is the boundary to the domain and $\mathbf{n}$ is the unit outward normal to this boundary.

$\mathbf{W}$ is the vector of dependent variables

$$\mathbf{W} = \begin{bmatrix} \rho \\ \rho U \\ \rho V \\ \rho E \\ \rho k \end{bmatrix} \qquad \ldots (2)$$

ρ, E and k are the density, total energy per unit mass and turbulent kinetic energy per unit mass respectively; U and V are the cartesian mean-velocity components. $\mathbf{H}$ is a matrix containing the flux vectors, which can be written as

$$\mathbf{H} = (\mathbf{F}^i + \mathbf{F}^v)\mathbf{i} + (\mathbf{G}^i + \mathbf{G}^v)\mathbf{j} \qquad \ldots (3)$$

where $\mathbf{i}$ and $\mathbf{j}$ are unit vectors in the X- and Y-directions of the cartesian coordinate system. $\mathbf{F}^i$ and $\mathbf{G}^i$ are the convective flux vectors

$$\mathbf{F}^i = \begin{bmatrix} \rho U \\ \rho U^2 + P \\ \rho UV \\ \rho UH \\ \rho Uk \end{bmatrix}, \mathbf{G}^i = \begin{bmatrix} \rho V \\ \rho UV \\ \rho V^2 + P \\ \rho VH \\ \rho Vk \end{bmatrix} \qquad \ldots (4)$$

P is the static pressure and H the total enthalpy per unit mass. The various quantities are related to each other by the definitions of total energy per unit volume and total enthalpy per unit volume for a perfect gas

$$\rho E = P/(\gamma - 1) + \tfrac{1}{2}\rho(U^2 + V^2) + \rho k \qquad \ldots (5)$$

$$\rho H = \rho E + P \qquad \ldots (6)$$

γ being the ratio of specific heats.

$\mathbf{F}^v$ and $\mathbf{G}^v$ are the viscous diffusive flux vectors

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \sigma_{xx} \\ \sigma_{xy} \\ U\sigma_{xx} + V\sigma_{xy} + q_x \\ \mathrm{diff}(k)_x \end{bmatrix}$$

$$\mathbf{G}^v = \begin{bmatrix} 0 \\ \sigma_{xy} \\ \sigma_{yy} \\ U\sigma_{xy} + V\sigma_{yy} + q_y \\ \mathrm{diff}(k)_y \end{bmatrix} \qquad \ldots (7)$$

σ_{xx}, σ_{yy} and σ_{xy} are components of the stress tensor, whilst q_x and q_y are components of the heat-flux vector. $\mathrm{diff}(k)_x$ and $\mathrm{diff}(k)_y$ represent viscous diffusion in the transport equation for the turbulent kinetic energy, and the exact form adopted for these terms is discussed in Section 7 below. Finally, vector $\mathbf{S}^v$ contains the source terms in the modelled transport equation for k

$$\mathbf{S}^v = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ S_k \end{bmatrix} \qquad \ldots (8)$$

where S_k is defined in Equation (58) below. Reynolds stresses and turbulent heat fluxes in the mean-flow equations are modelled by introducing an isotropic eddy viscosity μ_t and a turbulent Prandtl number Pr_t. Thus, the viscous stress terms in Equation (7) become

$$\sigma_{xx} = -(\mu + \mu_t)\, s_{xx} + \frac{2}{3}\rho k,$$

$$\sigma_{yy} = -(\mu + \mu_t)\, s_{yy} + \frac{2}{3}\rho k,$$

$$\sigma_{xy} = -(\mu + \mu_t)\, s_{xy} \qquad \ldots (9)$$

where the components of the mean-strain tensor are

$$s_{xx} = 2\frac{\partial U}{\partial X} - \frac{2}{3}\left(\frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y}\right),$$

$$s_{yy} = 2\frac{\partial V}{\partial Y} - \frac{2}{3}\left(\frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y}\right),$$

$$s_{xy} = \left(\frac{\partial U}{\partial Y} + \frac{\partial V}{\partial X}\right) \qquad \ldots (10)$$

Similarly, the components of the heat-flux vector in Equation (7) become

$$q_x = -\gamma\,(\mu/\mathrm{Pr} + \mu_t/\mathrm{Pr}_t)\,\partial T/\partial X,$$

$$q_y = -\gamma\,(\mu/\mathrm{Pr} + \mu_t/\mathrm{Pr}_t)\,\partial T/\partial Y \quad . . . \quad (11)$$

Pr and Pr_t, the laminar and turbulent Prandtl numbers, are assumed to take constant values of 0·72 and 0·9 respectively. The static temperature T in Equation (11) can be evaluated using the perfect gas relation, written in the form

$$P = (\gamma - 1)\,\rho\,T \quad . . . \quad (12)$$

the pressure P being determined in terms of the dependent variables in $\mathbf{W}$, using the definition of the total energy, Equation (5). Sutherland's law

$$\mu = \frac{\gamma^{1/2}\mathrm{M}}{\mathrm{R}}[(\gamma - 1)\,T]^{3/2}\left[\frac{T^*_\infty + 110{\cdot}4}{(\gamma - 1)\cdot T\,T^*_\infty + 110{\cdot}4}\right] \quad . . . \quad (13)$$

is used to evaluate the laminar viscosity μ, where M and R are the freestream Mach number and Reynolds number respectively. T^*_∞ is the freestream static temperature, in kelvin.

The surface boundary conditions for Equation (1) are the no-slip conditions

$$U_w = V_w = k_w = 0 \quad . . . \quad (14)$$

together with the assumption of an adiabatic wall, which leads to

$$(\partial T/\partial y_n)_w = (\partial P/\partial y_n)_w = 0 \quad . . . \quad (15)$$

Subscript w indicates conditions at the wall and y_n is the surface normal distance.

Non-reflecting farfield boundary conditions are applied at the outer boundary to the computational domain. These are constructed using the Riemann invariants for a one-dimensional flow normal to the outer boundary; *see* Jameson and Baker[16] for details. The downstream outflow boundary in the wake region is treated in exactly the same way. For subsonic outflow, this amounts to an extrapolation from the interior of the entropy and the mean-velocity component tangential to the downstream boundary. The normal mean-velocity component and the speed of sound are given by the Riemann invariants, and the pressure is determined using the definition of the speed of sound, Equation (40). For supersonic outflow, all quantities are extrapolated from the interior.

The following scalings have been adopted to write the governing flow equations, and all subsequent equations, in non-dimensional form

$$X^* = X\,c^*,\; Y^* = Y\,c^*,\; \rho^* = \rho\,\rho^*_\infty,\; P^* = P\,P^*_\infty,$$

$$U^* = U\sqrt{(P^*_\infty/\rho^*_\infty)},\; V^* = V\sqrt{(P^*_\infty/\rho^*_\infty)},$$

$$E^* = E\,P^*_\infty/\rho^*_\infty,\; H^* = H\,P^*_\infty/\rho^*_\infty,\; k^* = k\,P^*_\infty/\rho^*_\infty,$$

$$T^* = T\,P^*_\infty/(\rho^*_\infty\,C^*_v),\; t^* = t\,c^*/\sqrt{(P^*_\infty/\rho^*_\infty)},$$

$$\mu^* = \mu\,c^*\sqrt{(P^*_\infty\,\rho^*_\infty)},\; \mu^*_t = \mu_t\,c^*\sqrt{(P^*_\infty\,\rho^*_\infty)} \quad . . . \quad (16)$$

Superscript * indicates a dimensional quantity and subscript ∞ denotes freestream conditions. C^*_v is the specific heat at constant volume, and the reference length scale c^* is taken as the chord of the aerofoil.

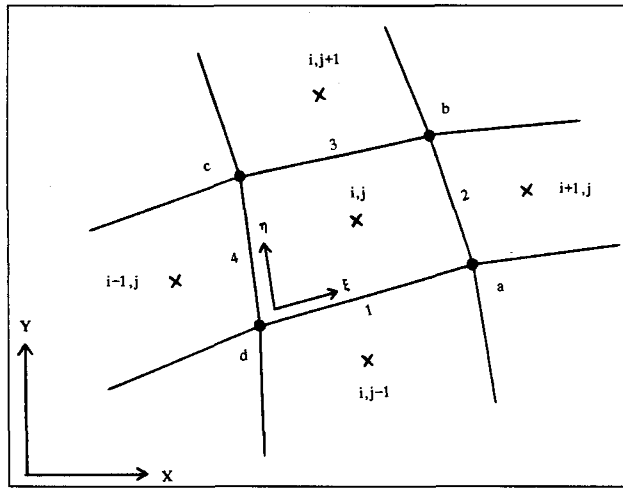

Figure 1. Notation for a typical computational cell, x = cell centres; ● = cell vertices.

4. SPATIAL DISCRETISATION

The computational domain Ω is divided into a finite number of non-overlapping quadrilateral cells, the resulting grid being boundary-conforming. Figure 1 shows a typical cell, which has four edges numbered 1 to 4 and four vertices (a, b, c, d). It is convenient at this point to define a local curvilinear coordinate system (ξ, η), for use when constructing the artificial dissipative terms; see Section 5 below. Note, however, that the governing flow equations retain their cartesian form. The dependent variables, Equation (2), within a cell are represented by their average values at the cell centre, such quantities being denoted by suffices (i, j) in the two curvilinear coordinate directions.

The integral conservation equations are now applied to each computational cell in turn. Equation (1) for a typical cell becomes

$$\frac{\partial \mathbf{W}_{i,j}}{\partial t} = -\frac{1}{h_{i,j}}\int_{\Omega_c}[(\mathbf{F}^i + \mathbf{F}^v)\mathrm{d}Y - (\mathbf{G}^i + \mathbf{G}^v)\,\mathrm{d}X] - \mathbf{S}^v_{i,j} \quad . . . \quad (17)$$

where Ω_c refers to a contour integration around the boundary of the cell, taken in the anticlockwise sense. The cell has an area of $h_{i,j}$ which is fixed in time.

The time-dependent integral form of these equations suggests performing the spatial discretisation using a finite-volume technique, and then marching the resulting semi-discrete equations in time to reach the steady-state solution. Performing the spatial discretisation of Equation (17) before the time discretisation leads to a large set of ordinary differential equations with respect to time

$$\frac{\mathrm{d}\mathbf{W}_{i,j}}{\mathrm{d}t} = -(\mathbf{Q}_{i,j} + \mathbf{V}_{i,j}) \quad . . . \quad (18)$$

$\mathbf{Q}_{i,j}$ and $\mathbf{V}_{i,j}$ involve discrete approximations of the inviscid and viscous flux integrals, the latter also containing the source terms $\mathbf{S}^v_{i,j}$.

The contour integration of the inviscid flux vectors, Equation (4), is approximated by

$$\mathbf{Q}_{i,j} = \frac{1}{h_{i,j}}\sum_{k=1}^{4}(\mathbf{F}^i_k\,\Delta Y_k - \mathbf{G}^i_k\,\Delta X_k) \quad . . . \quad (19)$$

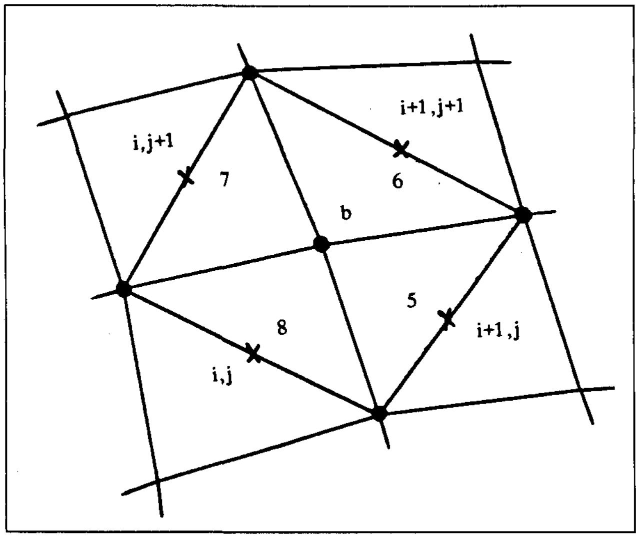

Figure 2. Typical auxiliary cell, for viscous flux computation.

The summation is over the four edges of the cell, Fig. 1, and

$$\Delta X_1 = X_a - X_d, \Delta X_2 = X_b - X_a,$$

$$\Delta X_3 = X_c - X_b, \Delta X_4 = X_d - X_c \quad \ldots \quad (20)$$

ΔY_1 to ΔY_4 being defined in a similar manner. Cell-edge values of the inviscid flux terms are approximated by the average of the two adjacent cell-centre values, for example

$$\mathbf{F}_1^i = (\mathbf{F}_{i,j}^i + \mathbf{F}_{i,j-1}^i)/2 \quad \ldots \quad (21)$$

The same procedure is used to approximate the contour integration of the viscous flux vectors, Equation (7),

$$\mathbf{V}_{i,j} = \frac{1}{h_{i,j}} \sum_{k=1}^{4} (\mathbf{F}_k^v \Delta Y_k - \mathbf{G}_k^v \Delta X_k) + \mathbf{S}_{i,j}^v \quad \ldots \quad (22)$$

Again, cell-edge values of the viscous flux terms are approximated by the average of the two adjacent cell-centre values, as in Equation (21).

Evaluation of Equation (22) requires cell-edge values of first derivatives of U, V and T with respect to X and Y. Derivatives at the cell vertices (a, b, c, d) can be determined by considering auxiliary cells surrounding each vertex, Fig. 2. A discrete application of the divergence theorem to the auxiliary cell surrounding vertex b results in

$$(\partial U/\partial X)_b = \frac{1}{h_b} \sum_{k=5}^{8} U_k \Delta Y_k,$$

$$(\partial U/\partial Y)_b = -\frac{1}{h_b} \sum_{k=5}^{8} U_k \Delta X_k \quad \ldots \quad (23)$$

where h_b is the area of the auxiliary cell. ΔX_5 to ΔX_8 and ΔY_5 to ΔY_8 are defined in a similar manner to Equation (20). Variables at the edges of the auxiliary cell are approximated by the relevant cell-centre values, for example

$$U_5 = U_{i+1,j} \quad \ldots \quad (24)$$

Cell-edge values of the derivatives are then approximated by

$$(\partial U/\partial X)_1 = [(\partial U/\partial X)_a + (\partial U/\partial X)_d]/2 \quad \ldots \quad (25)$$

Finally, the source term in Equation (22) requires the evaluation of first derivatives of U and V at the cell centre, *see* Equations (58) and (59) below. Applying the divergence theorem to the main computational cell, Fig. 1, enables these derivatives to be approximated as

$$(\partial U/\partial X)_{i,j} = \frac{1}{h_{i,j}} \sum_{k=1}^{4} U_k \Delta Y_k,$$

$$(\partial U/\partial Y)_{i,j} = -\frac{1}{h_{i,j}} \sum_{k=1}^{4} U_k \Delta X_k \quad \ldots \quad (26)$$

5. ARTIFICIAL DISSIPATION

Cell-centred spatial discretisation schemes such as the one described above are non-dissipative, so that any errors (truncation, round-off etc) are not damped in time, and oscillations may be present in the steady-state solution. In order to eliminate these oscillations, artificial dissipative terms are added to Equation (18), which becomes

$$\frac{d\mathbf{W}_{i,j}}{dt} = -(\mathbf{Q}_{i,j} + \mathbf{V}_{i,j} - \mathbf{D}_{i,j}) \quad \ldots \quad (27)$$

The approach of Jameson *et al*[14] is adopted to construct the dissipation function $\mathbf{D}_{i,j}$ consisting of a blend of second and fourth differences of the conserved variables $\mathbf{W}_{i,j}$. Fourth differences are added everywhere in the flow domain where the solution is smooth, but are 'switched-off' in the region of shock waves. A term involving second differences is then 'switched-on' to damp oscillations near shock waves. This switching is achieved by means of a shock-wave sensor, based on the local second differences of pressure.

In detail, the artificial dissipative terms are constructed as follows

$$\mathbf{D}_{i,j} = (\mathbf{D}^\xi + \mathbf{D}^\eta)_{i,j} \quad \ldots \quad (28)$$

where $\mathbf{D}^\xi$ and $\mathbf{D}^\eta$ are components with respect to the curvilinear coordinates ξ and η, Fig. 1. Since the computational grid is structured, the cell centres are defined by the two indices (i, j) in these coordinate directions. The two components of Equation (28) are now written in terms of differences of cell-edge values.

$$\mathbf{D}_{i,j} = (\mathbf{d}_2 - \mathbf{d}_4 + \mathbf{d}_3 - \mathbf{d}_1)/\Delta t_{i,j}^* \quad \ldots \quad (29)$$

$\Delta t_{i,j}^*$ is the local cell-centre time step for a CFL number of unity, *see* Equation (42) below. The cell-edge components of the artificial dissipation are composed of first and third differences of the dependent variables, for example

$$\mathbf{d}_2 = \mathbf{d}_2^{(2)} - \mathbf{d}_2^{(4)} \quad \ldots \quad (30)$$

with

$$\mathbf{d}_2^{(2)} = \epsilon_2^{(2)} (\mathbf{W}_{i+1,j} - \mathbf{W}_{i,j}),$$

$$\mathbf{d}_2^{(4)} = \epsilon_2^{(4)} (\mathbf{W}_{i+2,j} - 3\mathbf{W}_{i+1,j} + 3\mathbf{W}_{i,j} - \mathbf{W}_{i-1,j}) \quad \ldots \quad (31)$$

where the differences are taken in the appropriate curvilinear direction. The adaptive coefficients

$$\epsilon_2^{(2)} = \kappa^{(2)} \max (\nu_{i+1,j}, \nu_{i,j}),$$

$$\epsilon_2^{(4)} = \max (0, \kappa^{(4)} - \epsilon_2^{(2)}) \qquad . . . (32)$$

are switched on or off by use of the shock-wave sensor ν, where

$$\nu_{i,j} = \frac{| P_{i+1,j} - 2P_{i,j} + P_{i-1,j} |}{| P_{i+1,j} + 2P_{i,j} + P_{i-1,j} |} \qquad . . . (33)$$

$\kappa^{(2)}$ and $\kappa^{(4)}$ are constants, taken equal to 0·25 and 0·004 respectively in the calculations presented below.

Note that the above approach to construction of the numerical dissipative terms follows closely that developed by Jameson *et al*[14] for use with the Euler equations. The present work indicates that this approach is also satisfactory for application to the Reynolds-averaged Navier–Stokes equations and the modelled transport equation for the turbulent kinetic energy. However, care must be taken to ensure that the artificial dissipative terms do not 'swamp' the real viscous effects in the boundary layers and wakes. Boundary layer theory indicates that it is generally gradients of mean-flow quantities across the viscous layers which dominate the flow development. Referring to Fig. 1, if the ξ-direction is identified as the surface tangential (or wake centreline) direction, mean-flow gradients in the η-direction will be dominant. Thus, if the differences of flow variables in the η-direction are used to construct the artifical dissipative terms, Equation (28), these may interfere with the natural flow development. The $\mathbf{D}^{\eta}$ terms in Equation (28) are deleted in order to prevent this behaviour. Experience indicates that such an approach results in artificial dissipative terms $\mathbf{D}_{i,j}$ which are at least an order of magnitude smaller than the real viscous terms $\mathbf{V}_{i,j}$ within the boundary layers and wakes.

6. TIME-MARCHING SCHEME

The spatial discretisation described above reduces the governing flow equations to a large system of semi-discrete ordinary differential equations, which may be written as

$$\frac{d\mathbf{W}_{i,j}}{dt} = \mathbf{R}_{i,j} \qquad . . . (34)$$

The right-hand side of the above system represents the residual error (deviation from the steady-state) at each cell centre (i, j)

$$\mathbf{R}_{i,j} = -(\mathbf{Q}_{i,j} + \mathbf{V}_{i,j} - \mathbf{D}_{i,j}) \qquad . . . (35)$$

The integration in time of Equation (34), to a steady-state solution, is performed using an explicit multi-stage scheme. Since time accuracy is not important for a steady-state solution, such schemes are selected only for their properties of stability region and damping. The following 4-stage scheme is adopted for the present work (neglecting, for clarity, the subscripts i, j denoting cell-centre quantities)

$$\mathbf{W}^{(0)} = \mathbf{W}^{n},$$

$$\mathbf{W}^{(m)} = \mathbf{W}^{(0)} + \alpha_m \Delta t\, \mathbf{R}^{(m-1)} \text{ for } m = 1 \text{ to } 4,$$

$$\mathbf{W}^{n+1} = \mathbf{W}^{(4)} \qquad . . . (36)$$

where n is the current time level, $n + 1$ is the new time level and

$$\mathbf{R}^{(m)} = -[\mathbf{Q}^{(m)} + \mathbf{V}^{(0)} - \mathbf{D}^{(0)}] \qquad . . . (37)$$

When solving the Euler equations for inviscid flow, the above 4-stage scheme is normally used in conjunction with the following coefficients

$$\alpha_1 = 1/4, \alpha_2 = 1/3, \alpha_3 = 1/2, \alpha_4 = 1 \qquad . . . (38a)$$

However, experience with the present method, solving the Reynolds-averaged Navier–Stokes equations for viscous flow, indicates that a modified set of coefficients

$$\alpha_1 = 0{\cdot}25, \alpha_2 = 0{\cdot}5, \alpha_3 = 0{\cdot}55, \alpha_4 = 1{\cdot}0 \qquad . . . (38b)$$

results in a more efficient time-marching procedure. In particular, improved damping is achieved of large wavelength oscillations that can appear in the surface pressure distribution during the transient phase, when the flow is locally close to sonic conditions. Jameson[17] uses these same coefficients when solving the Euler equations by a multi-grid procedure, citing better damping over a wide range of frequencies compared with the standard coefficients, Equation (38a).

In order to minimise the computation time, the expensive evaluation of the dissipation function $\mathbf{D}$ is carried out only at the first intermediate stage (0), and then frozen for the subsequent stages. This is known to modify the stability region of the scheme[17], but the steady-state accuracy and convergence characteristics are preserved. Similarly, the viscous flux and source terms comprising $\mathbf{V}$ are also only evaluated at stage (0) and then frozen for subsequent stages. These procedures do not appear to destabilise the time-marching procedure in any way. Numerically, the above scheme, with the coefficients given in Equation (38b), is found to have a maximum CFL number for stability of 2·0 when applied to the Reynolds-averaged Navier–Stokes equations. This represents a reduction from a value of about 2·6 possible with the standard coefficients, Equation (38a). However, the effect on the rate of convergence appears to be insignificant.

The major disadvantage of explicit schemes is that the magnitude of the maximum allowable time step is restricted because of the limited stability region. This restriction can be alleviated, to some extent, by marching the solution in each computational cell at the maximum allowable local time step. The transient solution is then not time-accurate, but this is of no concern for present purposes since only the final steady-state solution is of interest.

The local time step for a given computational cell is determined by the maximum velocities at which waves are propagated across the cell by convective and viscous diffusive processes. These velocities are related to the two maximum wave eigenvalues of the convective flux terms, Equation (4), in the mean-flow equations. It is convenient to define the resulting wave eigenvalues in terms of the local curvilinear coordinate system, Fig. 1,

$$\lambda_\xi^w = | Y_\eta U - X_\eta V | + a \sqrt{(X_\eta^2 + Y_\eta^2)}$$

$$\lambda_\eta^w = | X_\xi V - Y_\xi U | + a \sqrt{(X_\xi^2 + Y_\xi^2)} \qquad . . . (39)$$

where a is the local speed of sound

$$a^2 = \gamma P/\rho \qquad . . . (40)$$

and $X_\xi \equiv \partial X/\partial \xi$ etc. The viscous diffusive flux terms, Equation (7), have a damping effect on the wave propagation. Following Coakley[18], appropriate diffusion eigenvalues can be defined as

$$\lambda_\xi^d = 2\,[(\mu + \mu_t)/\rho]\,[(X_\eta^2 + Y_\eta^2)/h]$$

$$\lambda_\eta^d = 2\,[(\mu + \mu_t)/\rho]\,[(X_\xi^2 + Y_\xi^2)/h] \quad . \; . \; . \quad (41)$$

h being the cell volume. The local time step is evaluated as a combination of the four eigenvalues

$$\Delta t_{i,j}^* = h_{i,j}/\max\,[(\lambda_\xi^w + \lambda_\eta^w), (\lambda_\xi^d + \lambda_\eta^d)] \quad . \; . \; . \quad (42)$$

This time step is then multiplied by the CFL number for use in the multi-stage scheme

$$\Delta t_{i,j} = \mathrm{CFL}\;\Delta t_{i,j}^* \quad . \; . \; . \quad (43)$$

The rate of convergence to the steady-state can be further enhanced by using the implicit residual smoothing technique, as described by Jameson and Baker[16], and particularly the multi-grid procedure of Martinelli *et al*[19, 20]. These convergence acceleration techniques are under investigation for use with the present method, but have not been employed to obtain the results given below. This is in accordance with the philosophy, expressed in the introduction, of keeping the solution algorithm simple, in order to concentrate effort on turbulence model implementation.

7. TURBULENCE MODELLING

The use of mass-weighted average variables in compressible flows, together with modelling of Reynolds-stress terms via an eddy viscosity, results in a set of equations for the mean-flow development having a form very similar to the Navier–Stokes equations for viscous laminar flow. Turbulence modelling then essentially reduces to the specification of the local values of the eddy viscosity μ_t.

Two relatively simple turbulence models are considered in this section. The first is a pure algebraic model, and involves no additional transport equations for turbulence quantities. As such, the model assumes local equilibrium conditions, with turbulence being created and destroyed at the same position. The model is not expected to perform well in situations where turbulence history effects are important. The second model involves a transport equation for the turbulent kinetic energy, with terms representing convective and diffusive transport, production and dissipation (i.e. destruction). In principle, this model incorporates more realistic physics, and so should lead to improved results for flows significantly out-of-equilibrium.

7.1 Baldwin–Lomax Model

This is a purely algebraic turbulence model, whose formulation is based on the boundary-layer model of Cebeci and Smith[21], the Baldwin–Lomax model being more suited to implementation in solvers for the Reynolds-averaged Navier–Stokes equations. The present formulation of the model differs in detailed aspects from that described originally by Baldwin and Lomax[15], reflecting some of the practical implementation problems encountered (such problems are by no means unique to the present flow solver).

The eddy-viscosity distribution for the wall boundary layer is considered first, and this is defined in two parts. A mixing length formulation is used in the inner near-wall region

$$\mu_{ti} = \rho\, l_m^2\, |\,\omega\,| \quad . \; . \; . \quad (44)$$

where ω is the vorticity

$$\omega = \partial U/\partial Y - \partial V/\partial X \quad . \; . \; . \quad (45)$$

and l_m is the mixing length

$$l_m = [1 - \exp(-y^+/A^+)]\,\kappa\, y_n \quad . \; . \; . \quad (46)$$

y_n is the surface normal distance, and y^+ is defined as

$$y^+ = \rho_w\,|\,U_\tau\,|\,y_n/\mu_w \quad . \; . \; . \quad (47)$$

Subscript w denotes conditions at the wall. U_τ is the wall friction velocity and is determined from the wall shear stress τ_w

$$\tau_w = \rho_w\, U_\tau^2 = \mu_w\,(\partial U_T/\partial y_n)_w \quad . \; . \; . \quad (48)$$

U_T is the mean-velocity component tangential to the surface.

The outer eddy-viscosity formulation is given by

$$\mu_{to} = C_{cs}\, C_{cp}\, \rho\, F_{\mathrm{wake}}/[1 + 5{\cdot}5(C_{kleb} y_n/y_{\max})^6] \quad . \; . \; . \quad (49)$$

where F_{wake} is determined from

$$\left.\begin{aligned} F_{\mathrm{wake}} &= y_{\max}\, F_{\max} \\ &= C_{\mathrm{wk}}\, y_{\max}\, U_{\mathrm{dif}}^2/F_{\max} \end{aligned}\right\} \text{ smaller of these} \quad . \; . \; . \quad (50)$$

$F_{\max}$ is the maximum value of the function

$$F = [1 - \exp(-y^+/A^+)]\, y_n\,|\,\omega\,| \quad . \; . \; . \quad (51)$$

$y_{\max}$ being the value of y_n *for* $F_{\max}$. U_{dif} is the maximum velocity difference across the boundary layer

$$U_{\mathrm{dif}} = \sqrt{(U^2 + V^2)_{\max}} - \sqrt{(U^2 + V^2)_{\min}} \quad . \; . \; . \quad (52)$$

The turbulence model involves the following six constants

$$\kappa = 0{\cdot}41,\ A^+ = 26,\ C_{cs} = 0{\cdot}0168,$$

$$C_{cp} = 1{\cdot}6,\ C_{wk} = 0{\cdot}25,\ C_{kleb} = 0{\cdot}3 \quad . \; . \; . \quad (53)$$

A blending function is applied to the inner and outer formulations to ensure a smooth eddy-viscosity distribution across the boundary layer

$$\mu_t = \mu_{to}\tanh\,(\mu_{ti}/\mu_{to}) \quad . \; . \; . \quad (54)$$

There are a number of practical implementation problems associated with the Baldwin–Lomax turbulence model. The first concerns the determination of $y_{\max}$, which involves a search across the boundary layer at each station to find the position of $F_{\max}$, the maximum value of the function F, Equation (51). The length scale $y_{\max}$ is used as a characteristic scale for the outer part of the boundary layer, and so near-wall peaks in F must be ignored. This is accomplished by restricting the search for $F_{\max}$ to $y^+ > 10$. Secondly, multiple peaks in F may occur, particularly downstream of shock waves. An effective strategy is to search across the boundary layer, starting at the wall, for the first three peaks and to take the largest of these peaks to determine $y_{\max}$.

Evaluation of the outer velocity scale U_{dif}, Equation (52), requires the edge of the boundary layer to be found. Stock and Haase[22] propose scaling the boundary layer thickness δ with $y_{\max}$

$$\delta = 1{\cdot}936\, y_{\max} \quad . \; . \; . \quad (55)$$

this relationship being derived on the basis of a two-parameter velocity profile family for turbulent boundary layers. Equation (55) is used in the present method to evaluate δ.

A related study by Granville[23] indicates that C_{cp} and C_{kleb}, Equation (53), are not really constants and are not independent of each other. He presents a functional relationship between C_{cp} and C_{kleb}, and correlates C_{kleb} with a pressure gradient parameter for equilibrium boundary layers. Since flows involving shock wave/boundary layer interactions are unlikely to be in equilibrium, these refinements to the Baldwin-Lomax model are not implemented at present.

The final problem area concerns implementation of the turbulence model in wake regions. Baldwin and Lomax[15] suggest using only the outer eddy-viscosity formulation, but this can cause numerical problems due to the switch from Equation (54) to Equation (49) across the trailing edge. Alternatively, the inner formulation Equation (44), can be gradually relaxed away downstream of the trailing edge. There then remains the problem of a discontinuity in eddy viscosity across the wake centreline when the outer formulation is applied separately to the two half-wakes.

In view of these practical implementation problems, a rather crude wake model is adopted. The upper and lower surface trailing edge eddy-viscosity distributions are simply imposed at all stations downstream in the wake

$$(\mu_t)_{\text{wake}} = (\mu_t)_{\text{t.e.}} \qquad \ldots \quad (56)$$

This approach ensures continuity of the eddy viscosity across the trailing edge, alleviating numerical problems in that region. It is suggested that Equation (56) represents a reasonable approximation for the immediate near-wake region, which is the most important region as far as the flow development around the aerofoil is concerned.

7.2 One-Equation Model

The present one-equation turbulence model is very similar to that adopted by Mitcheltree *et al*[24]. The near-wall formulation of this latter model is a straightforward compressible flow adaptation of an incompressible flow model due to Wolfshtein[25]; see also Chen and Patel[26]. A modelled transport equation for the turbulent kinetic energy k is solved in conjunction with the mean-flow equations. Turbulent length scales, required to close the turbulence model, are defined in algebraic form. Thus, the one-equation model represents a 'halfway house' between a pure algebraic model, such as that of Baldwin and Lomax, and the more sophisticated two-equation models. (In fact, experience to date with two-equation turbulence models does not indicate any significant improvement in predictive capability for transonic aerofoil flows[11].)

The modelled k transport equation has already been given in outline in Section 3 above. The viscous diffusion terms in Equation (7) are modelled by assuming a simple scalar gradient diffusion process

$$\begin{aligned} \mathrm{diff}(k)_x &= -(\mu + \mu_t/\sigma_k)\frac{\partial k}{\partial X} \\ \mathrm{diff}(k)_y &= -(\mu + \mu_t/\sigma_k)\frac{\partial k}{\partial Y} \end{aligned} \qquad \ldots \quad (57)$$

Mitcheltree *et al*[24] use a more elaborate tensoral gradient diffusion model involving the Reynolds stresses. It is not apparent that this latter approach will result in improved physical modelling since the Reynolds stresses are still approximated using an eddy-viscosity formulation.

The source term S_k in Equation (8) contains production terms P_k, together with the rate of turbulent kinetic energy dissipation ϵ

$$S_k = -P_k + \rho\,\epsilon \qquad \ldots \quad (58)$$

where

$$\begin{aligned} P_k = {} & (\mu_t\, s_{xx} - \frac{2}{3}\rho\, k)\frac{\partial U}{\partial X} \\ & + (\mu_t\, s_{yy} - \frac{2}{3}\rho\, k)\frac{\partial V}{\partial Y} + \mu_t\, s_{xy}\, s_{xy} \end{aligned} \qquad \ldots \quad (59)$$

s_{xx}, s_{yy} and s_{xy} are the components of the mean-strain tensor, and are defined in Equation (10). The eddy-viscosity relation used in conjunction with the transport equation of k is

$$\mu_t = c_\mu\, \rho\, k^{1/2} L_\mu \qquad \ldots \quad (60)$$

The rate of dissipation of k is modelled via a dissipation length scale L_ϵ

$$\rho\,\epsilon = \rho\, k^{3/2}/L_\epsilon \qquad \ldots \quad (61)$$

The two length scales L_μ and L_ϵ are defined by algebraic relations, their formulation being in two parts for the wall boundary layer. In the inner region, the length scales vary linearly with distance from the wall, modified by damping functions in the molecular-viscosity dominated region immediately adjacent to the surface

$$\begin{aligned} L_{\mu i} &= c_1\, y_n\, [1 - \exp(-R_k/A_\mu)] \\ L_{\epsilon i} &= c_1\, y_n\, [1 - \exp(-R_k/2c_1)] \end{aligned} \qquad \ldots \quad (62)$$

where R_k is a Reynolds number characteristic of the turbulence

$$R_k = \rho\, k^{1/2}\, y_n/\mu \qquad \ldots \quad (63)$$

Both length scales are taken to be constant in the outer region, scaling with the boundary layer thickness δ

$$L_{\mu o} = L_{\epsilon o} = 0{\cdot}085\ \delta/c_\mu^{3/4} \qquad \ldots \quad (64a)$$

The relationship between δ and $y_{\max}$ proposed by Stock and Haase[22], Equation (55), is used to replace δ, leading to

$$L_{\mu o} = L_{\epsilon o} = y_{\max} \qquad (64b)$$

Blending functions are used to ensure a smooth matching of the inner and outer formulations

$$\begin{aligned} L_\mu &= L_{\mu o} \tanh\,(L_{\mu i}/L_{\mu o}) \\ L_\epsilon &= L_{\epsilon o} \tanh\,(L_{\epsilon i}/L_{\epsilon o}) \end{aligned} \qquad \ldots \quad (65)$$

The one-equation model involves five constants

$$\begin{aligned} & \sigma_k = 1{\cdot}0,\ c_\mu = 0{\cdot}09,\ \kappa = 0{\cdot}41, \\ & A_\mu = 76,\ c_1 = \kappa/c_\mu^{3/4} \end{aligned} \qquad \ldots \quad (66)$$

As with the Baldwin–Lomax model, the practical implementation of the turbulence model in wake regions is a problem.

One approach is to use the outer formulations for the two length scales L_μ and L_ϵ, applied separately in the upper and lower half-wakes. However, this can lead to numerical problems in the wake region immediately downstream of the trailing edge, due to the abrupt switch from Equation (65) to Equation (64b) across the trailing edge. The two length scales will also be discontinuous across the wake centreline.

The approach taken in the present work is to impose the trailing edge eddy-viscosity and dissipation-rate distributions at all stations downstream in the wake

$$(\mu_t)_{\text{wake}} = (\mu_t)_{\text{t.e.}},\ (\rho\,\epsilon)_{\text{wake}} = (\rho\,\epsilon)_{\text{t.e.}} \qquad . \quad . \quad . \quad (67)$$

These quantities thus remain continuous across the trailing edge and the wake centreline. Physically, the flow is expected to relax away from the boundary-layer formulation, Equation (65), to some kind of wake formulation, over a small downstream distance. Equation (67), therefore, appears to be a reasonable model for the near-wake region immediately downstream of the trailing edge. Also, the far wake is expected to have little influence on the flow development around the aerofoil.

7.3 Turbulence Model Implementation

The Baldwin–Lomax algebraic and k–L_ϵ one-equation turbulence models are highly compatible with the present solution algorithm for the mean-flow equations. No implementation problems are encountered, beyond those already discussed in Sections 7.1 and 7.2 above. Computations with the two models are 'self-starting', with all mean-flow quantities initialised to freestream values and, for the one-equation model, prescribing a constant background level of turbulent kinetic energy. The flow is essentially impulsively started, by a sudden imposition of the boundary conditions, and time marches away from the initial conditions to a steady-state solution.

Eddy-viscosity based turbulence models involving turbulence-transport equations are only loosely coupled to the mean-flow equations, via the eddy viscosity itself. It is therefore possible to decouple the solution of these additional equations from the basic mean-flow solver, and such an approach is taken with a number of existing methods. This has the merit of allowing complete freedom to employ different solution algorithms for the mean-flow and turbulence-transport equations. However, splitting up the two sets of equations can lead to unnecessary duplication of computer code, and can lower the overall level of code vectorisation achievable.

The present flow algorithm, being explicit in nature, is particularly amenable to vectorisation in a straightforward manner. A wish to maintain a high level of vectorisation suggests a simple strategy for implementation of the desired range of turbulence models. The transport equation for the turbulent kinetic energy k is fully integrated into the set of governing flow equations, as described in Section 3. A 'null' equation for k is solved with the Baldwin-Lomax model, by initialising the turbulent kinetic energy to zero throughout the flowfield, and setting the source term S_k in Equation (8) to zero. Subsequently, during the time-marching procedure, 'zero' turbulent kinetic energy is convected and diffused around the flow domain, but there is no mechanism to create non-zero values of k.

Adopting the above approach to turbulence model implementation results in a flow algorithm which is mainly independent of the actual model being employed. Identical discretisation and solution procedures are used for the mean-flow and turbulence-transport equations. In fact, the technique has been further extended in the present method to include the k–ϵ two-equation model, which involves a second modelled transport equation for ϵ, the rate of dissipation of k. A 'null' equation for ϵ is solved when employing the Baldwin–Lomax or one-equation models, a total of six (four mean-flow and two turbulence-transport) equations being solved for all the models. There is, of course, a penalty in computation time associated with solving one or two 'null' equations, so that the total execution time is approximately the same for the three turbulence models. However, the computer code does achieve high vectorisation levels of 92% and 94% on an Amdahl VP 1100, for the Baldwin–Lomax and one-equation models respectively. The corresponding CPU times are $2{\cdot}42 \times 10^{-5}$ and $2{\cdot}49 \times 10^{-5}$ seconds/computational cell/time step (Results for the k–ϵ model are to be presented at a later date).

The final implementation topic to be addressed concerns the treatment of laminar/turbulent transition. Free transition with the Baldwin–Lomax model can be simulated in a rather crude sense, by allowing only eddy viscosity levels above a certain threshold. Baldwin and Lomax[15] themselves suggest setting the eddy viscosity to zero at any stations for which the maximum computed value of μ_t across the viscous layer is less than a constant multiple of the molecular viscosity, $(\mu_t)_{\max} < 14\mu$ for example. In practice, the position of transition tends to 'wander around' using this approach, and can inhibit convergence of the time-marching scheme. Fixed transition points are therefore assumed in the calculations presented below, with the eddy viscosity being set to zero at all stations upstream of these points.

The one-equation model can, in principle, mimic the transition process, since the near-wall damping functions introduced to suppress turbulence as the wall is approached will also tend to reduce turbulence levels in low Reynolds number regions. Again, experience with free transition calculations indicates inhibition of convergence due to movement of transition position during the time-marching procedure. The treatment of laminar/turbulent transition with the one-equation model requires further investigation, and fixed transition has been assumed in calculations to date. This is achieved by setting both the eddy viscosity and the source term S_k to zero at stations upstream of the transition points.

8. RESULTS

Results are presented in this section of calculations for four different aerofoils, covering a range of conditions from fully subcritical flow to supercritical flow with shock-induced separation. The calculations are made using the method in a 'production mode', with a fixed set of flow algorithm and grid generation parameters. Thus, the multi-stage time-marching scheme uses the set of coefficients given in Equation (38b), with a CFL number of 2·0 and employs local time-stepping for a total of 8000 time steps. Constant values of 0·25 and 0·004 are used for the two artificial dissipation coefficients $\kappa^{(2)}$ and $\kappa^{(4)}$, Equation (32). In addition, the $\mathbf{D}^{\eta}$ contribution to the artificial dissipation, Equation (28), involving differences of flow variables across boundary layers and wakes, is set to zero.

Computational grids are constructed using the algebraic grid generator of Rizzi[27], which produces a grid of so-called C-type topology. The grid is then artificially split into three parts or blocks, Fig. 3, since the present application of the method uses a block-structured grid formulation (this is to

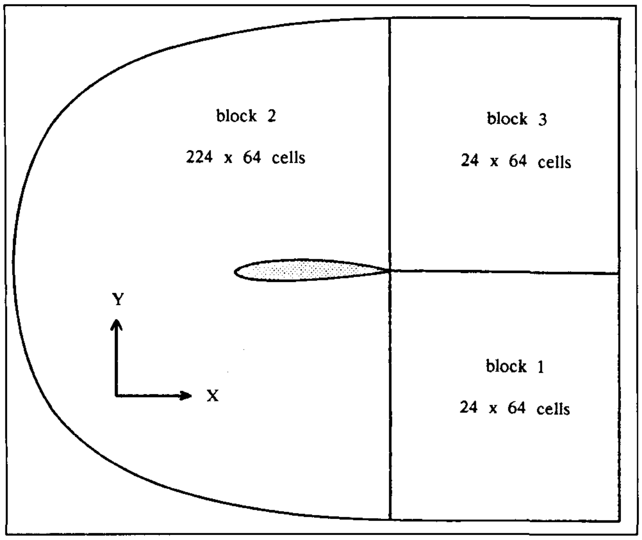

Figure 3. 3-block C-grid structure of computational grid.

enable the method to be applied, at a later date, to the multi-element aerofoil sections associated with mechanical high-lift systems). The grids for all four aerofoils are generated with a fixed set of parameters. Identifying the local curvilinear coordinates (ξ, η) as being in the surface tangential and normal directions respectively, block 2 around the aerofoil has dimensions of 224 × 64 cells in these two directions. Initial spacings $\Delta\xi/c$ and $\Delta\eta/c$ of 0·002 and 0·00005 are used in the leading- and trailing-edge regions, the latter spacing corresponding to a maximum y^+ value of about 5 at the cell centre adjacent to the surface. Such a near-wall value of y^+ is sufficiently small to enable an accurate numerical determination of the skin friction. Blocks 1 and 3 in the wake region have dimensions of 24 × 64 cells in the two curvilinear directions. The outer boundary to the computational domain is 15 chords away from the aerofoil surface and 10 chords downstream of the trailing edge. The inner regions of the resulting grids for the four aerofoils are shown in Fig. 4.

8.1 RAE 2822 Aerofoil

The RAE 2822 aerofoil has a maximum thickness/chord ratio of 12·1% and a sharp trailing edge. Cook *et al*[28] present an extensive experimental study of this aerofoil in the 8ft × 6ft transonic wind tunnel at RAE Farnborough. The data-set includes surface pressure, skin friction and integral thickness distributions as well as mean-velocity profiles, at a range of flow conditions. As such, it is one of the more nearly complete aerofoil data-sets and has been used extensively to validate numerical methods[11]. The aerofoil model has an aspect ratio of 3, and so the wind tunnel sidewalls should have little influence on the flow development. Similarly, the tunnel height/aerofoil chord ratio of 4 is large enough for linearised theory to be used to determine wind tunnel corrections for the upper and lower walls.

Figure 4(a) shows the RAE 2822 aerofoil with a close-up of the computational grid used in the present calculations. Experience indicates that the manufactured rather than the design geometry must be used, in order to capture correctly features of the surface pressure distribution in the leading-edge region. Four cases are considered here and the relevant flow conditions are given in Table 1(a). Comparisons of predicted and measured lift, drag and pitching moment coefficients are shown in Table 1(b), results being presented for both the Baldwin–Lomax and k–L_ϵ one-equation turbu-

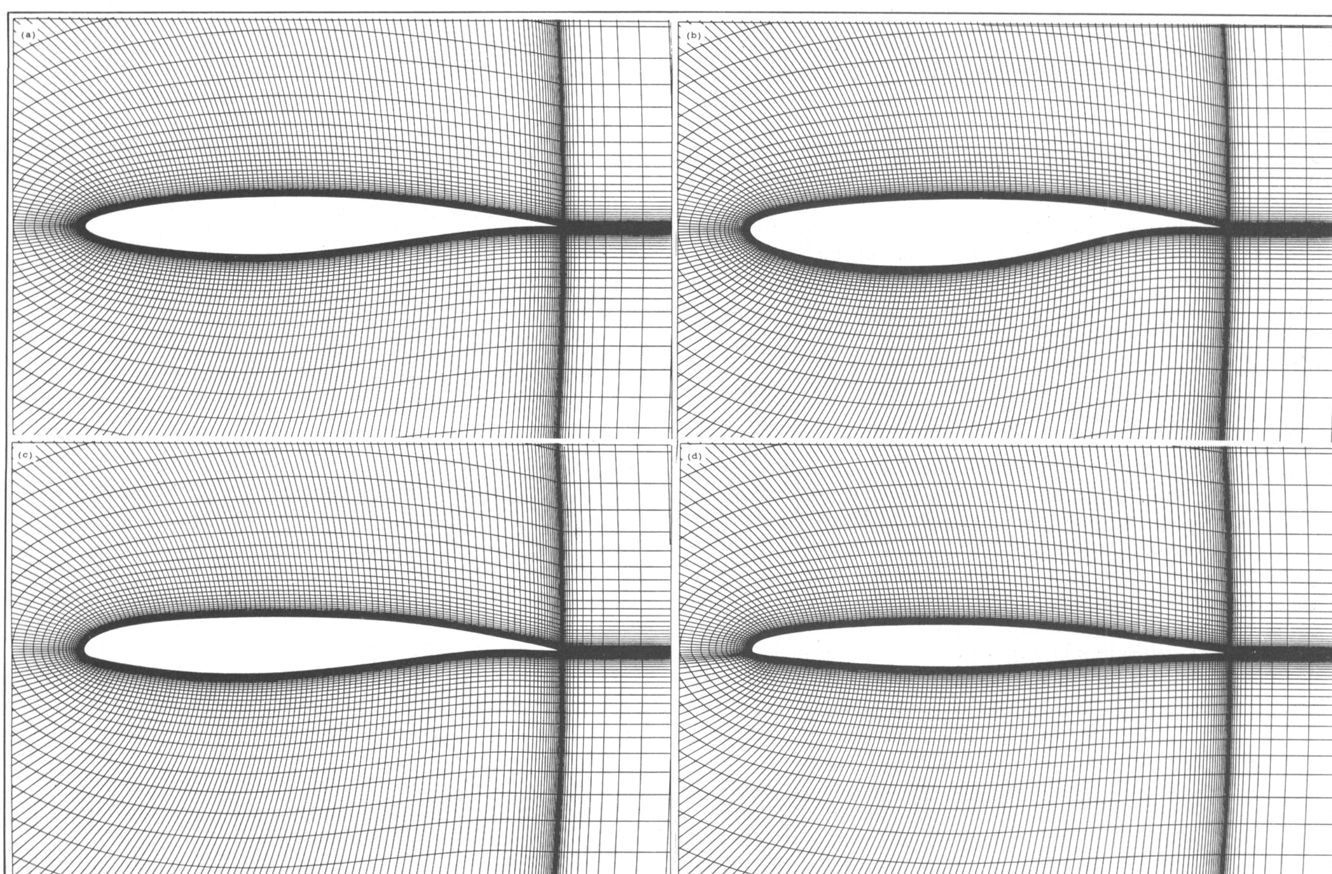

Figure 4. Aerofoil sections, including inner region of grids:
(a) RAE 2822 aerofoil; (b) RAE 5225 aerofoil; (c) CAST 7 aerofoil; (d) MBB-A3 aerofoil.

TABLE 1
RAE 2822 Aerofoil
Results using Baldwin–Lomax (B–L) and k–L_ϵ models

(a) Flow conditions

Case	M	α_{corr}	R × 10^{-6}
1	0·676	1·93°	5·7
6	0·725	2·54°	6·5
9	0·730	2·79°	6·5
10	0·750	2·81°	6·2

(b) Comparison of measured and predicted loads

Case 1	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·566	—	—	0·0085	−0·082
B–L	0·5802	0·004586	0·006093	0·01068	−0·08769
k–L_ϵ	0·5799	0·004486	0·006179	0·01067	−0·08752

Case 6	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·743	—	—	0·0127	−0·095
B–L	0·7457	0·009078	0·005937	0·01502	−0·09175
k–L_ϵ	0·7446	0·008908	0·006088	0·01500	−0·09145

Case 9	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·803	—	—	0·0168	−0·099
B–L	0·7950	0·01308	0·005760	0·01884	−0·09513
k–L_ϵ	0·7909	0·01275	0·005942	0·01869	−0·09407

Case 10	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·743	—	—	0·0242	−0·106
B–L	0·7901	0·02369	0·005521	0·02921	−0·1108
k–L_ϵ	0·7910	0·02247	0·005887	0·02836	−0·1095

lence models. Note that the measured incidence angles, lift and pitching moment coefficients in Table 1 include the relevant wind tunnel corrections, as described by Cook *et al*[28]. Transition is fixed at 11% chord for Case 1 and at 3% chord for Cases 6, 9 and 10, on the upper and lower surfaces of the aerofoil.

Figure 5 shows the results for Case 1 which involves mainly subcritical flow, apart from in the immediate leading-edge region. The predictions from the two turbulence models are virtually indistinguishable, and the agreement with measurements is generally very good. There are some detailed differences in the shape of the mean-velocity profiles, in the adverse pressure gradient region towards the rear of the upper surface, but these are relatively minor.

Case 6 Fig. 6, has a region of supercritical flow on the upper surface, terminated by a shock wave of moderate strength just downstream of the mid-chord position. Again, good agreement with experiment is obtained for most parameters, apart from in the immediate region of the shock wave which is predicted a little too far upstream. The reason for this mis-match in predicted and measured shock wave position is unclear at present, but is also found with other calculation methods; *see* Holst[11], Coakley[13], Martinelli and Jameson[20] for example (These authors all use the design aerofoil geometry, however). A closer study of the surface pressure distribution, Fig. 6(a), shows an overprediction of the strength of the shock wave, presumably due to an underprediction of the shock wave/boundary layer interaction. There results an underprediction of velocity deficit in the boundary layer downstream of the shock wave, but the predictions are recovering towards experiment as the trailing edge is approached.

Figure 7 shows that the results for Case 9, at a slightly higher freestream Mach number and incidence angle, are similar to those for Case 6, although there is better agreement in shock wave position. Differences between predictions and measurements are again confined to the immediate vicinity of the shock wave and in the downstream adverse pressure gradient region. The upper surface skin friction distributions, Fig. 7(c), indicate flow separation at the foot of the shock wave and close to the trailing edge for the Baldwin–Lomax turbulence model. No separation in the experiment is mentioned for this case[28], in agreement with the k–L_ϵ model results which predict fully-attached flow. This tendency of the Baldwin–Lomax model to predict premature flow separation, particularly at shock waves, has been noted by other authors[11]. The likely cause is the use of the friction velocity U_τ in the near-wall damping function for the inner eddy-viscosity formulation, Equation (46). In contrast, the k–L_ϵ model uses the square-root of the turbulent kinetic energy as the relevant inner velocity scale, Equation (62), a quantity which remains well-behaved (and positive) in separated flow regions.

Case 10 is the most interesting of the RAE 2822 cases to be considered, since it is the only one for which shock-induced separation is present in the experiment. Figure 8 shows that there are now larger discrepancies between predictions and measurements, and also between the results for the two turbulence models. In particular, Fig. 8(b) indicates problems with convergence when using the Baldwin–Lomax model, and this can be traced to the behaviour of the predictions downstream of the shock wave. Both turbulence models predict shock-induced separation, but the flow remains separated right to the trailing edge for the Baldwin–Lomax model. This behaviour is again attributed to the use of the friction velocity U_τ in the near-wall damping function.

Oil-flow visualisation in the experiment shows separated flow downstream of the shock wave, from an X/c of about 0·62 to 0·72. The k–L_ϵ model predicts a separated-flow region of slightly smaller extent but in the same position, Fig. 8(c). However, both turbulence models position the shock wave too far downstream compared to experiment, and the subsequent upper surface pressure levels are incorrect, Fig. 8(a). The mean-velocity profiles for the Baldwin–Lomax model show a slightly better response downstream of the shock wave, Fig. 8(g), but these results must be treated with caution since the solution is not fully-converged in this region. The main conclusion from Case 10 is that the shock wave/boundary layer interaction is significantly underpredicted in the presence of shock-induced separation, and this leads to further discrepancies in the downstream region.

8.2 RAE 5225 Aerofoil

The RAE 5225 aerofoil is a supercritical section of 14% maximum thickness with a sharp trailing edge, and has more rear loading than the RAE 2822 aerofoil. It has been extensively tested in the 8ft × 8ft transonic wind tunnel at RAE Bedford, at Reynolds numbers of 6 and 20 million; *see* Ashill *et al*[10] and also Lock[29]. The aerofoil model has an aspect ratio of 3·84, and so the tunnel sidewalls are expected to have little effect on the flow development. The tunnel walls are solid, with a height/aerofoil chord ratio of 3·84, allowing blockage and upwash effects to be defined precisely. Tunnel blockage is corrected by a simple Mach number increment, but Ashill *et al*[10] indicate that an incidence correction alone is not sufficient for the wall-induced upwash, and they use a camberline correction in BVGK calculations. This latter correction is found to have little effect in the present calculations and so is omitted.

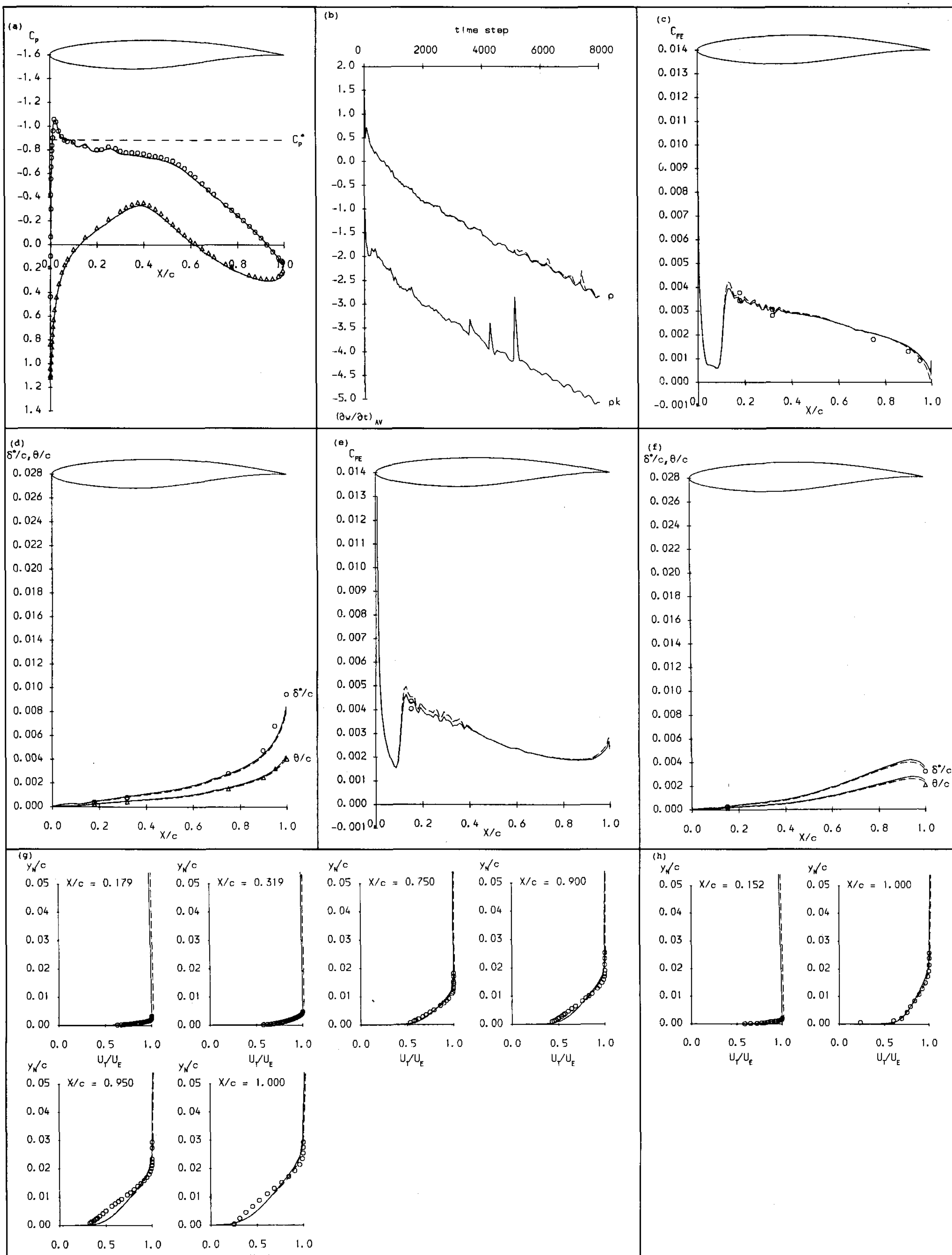

Figure 5. Results for RAE 2822 — case 1 — M = 0·676, α = 1·93° O, Δ experiment; ——— k–L_ϵ; — — — — Baldwin–Lomax: (a) surface pressure coefficient; (b) convergence history; (c) upper surface skin friction; (d) upper surface integral thicknesses; (e) lower surface skin friction; (f) lower surface integral thicknesses; (g) upper surface mean-velocity profiles; (h) lower surface mean-velocity profiles.

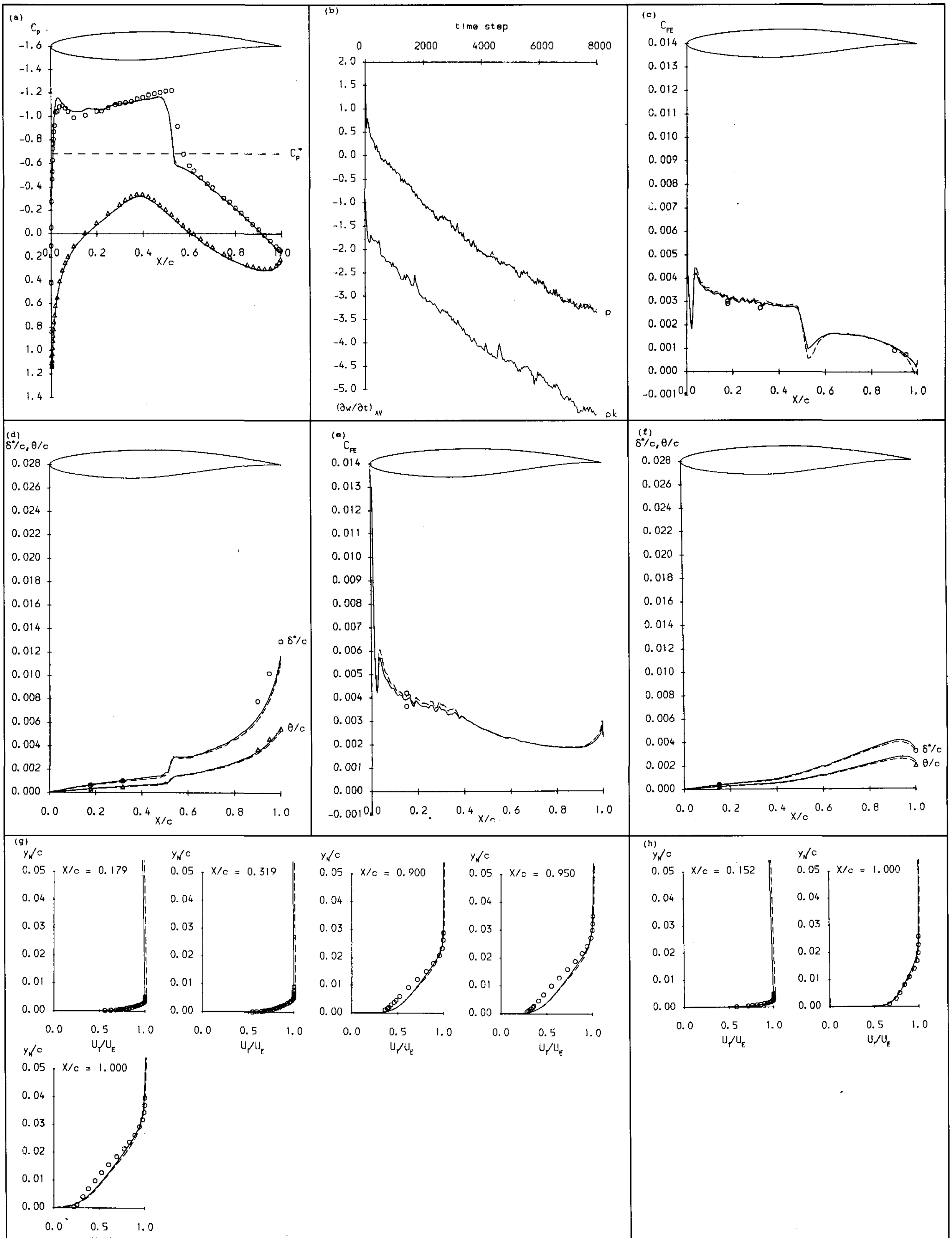

Figure 6. Results for RAE 2822 — case 6 — M = 0·725, α = 2·54° O, Δ experiment; ———— k–L_ϵ; — — — — Baldwin–Lomax: (a) surface pressure coefficient; (b) convergence history; (c) upper surface skin friction; (d) upper surface integral thicknesses; (e) lower surface skin friction; (f) lower surface integral thicknesses; (g) upper surface mean-velocity profiles; (h) lower surface mean-velocity profiles.

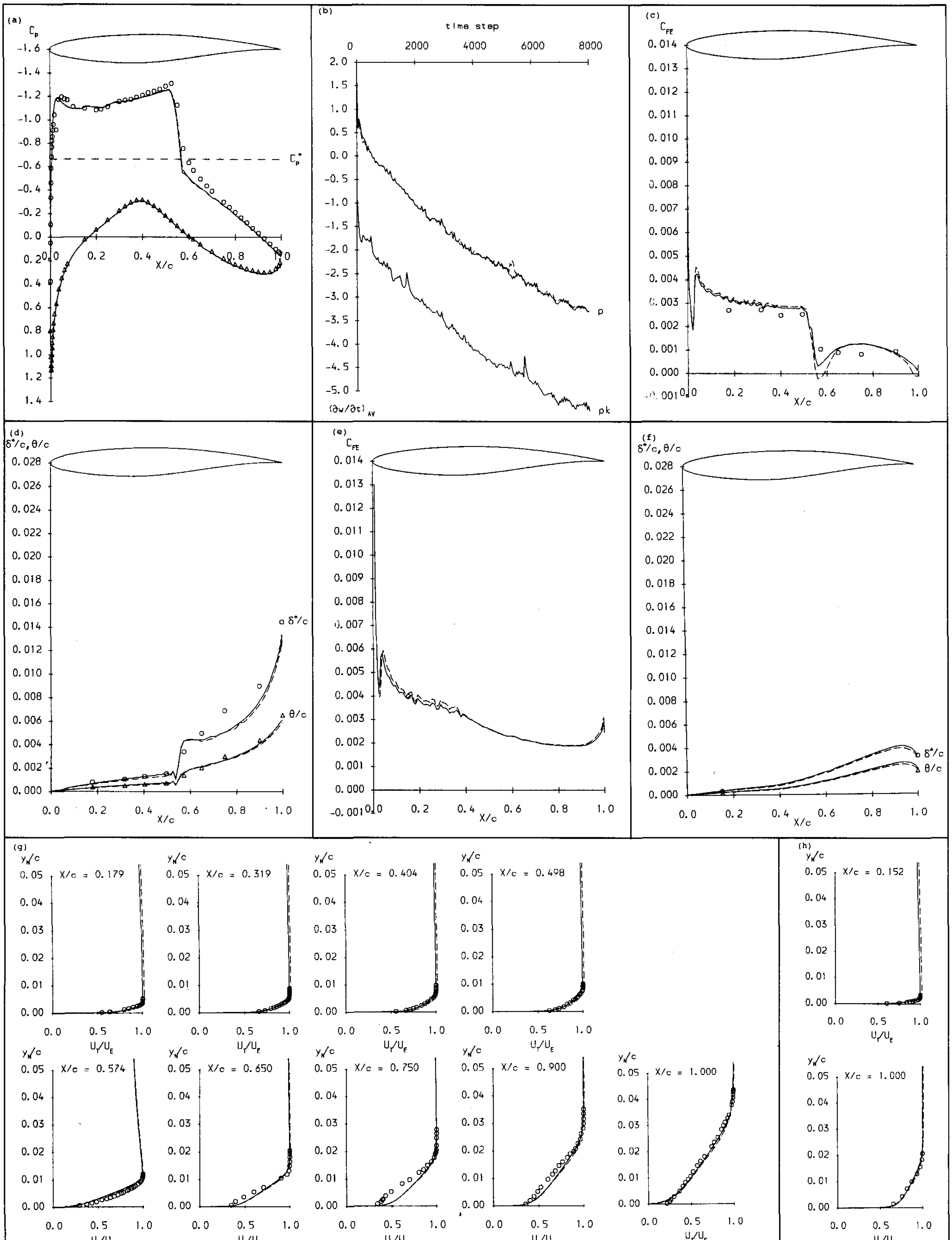

Figure 7. Results for RAE 2822 — case 9 — M = 0·73, α = 2·79° O, Δ experiment; ———— k–L_ϵ; — — — — Baldwin–Lomax:
(a) surface pressure coefficient; (b) convergence history; (c) upper surface skin friction; (d) upper surface integral thicknesses;
(e) lower surface skin friction; (f) lower surface integral thicknesses;
(g) upper surface mean-velocity profiles; (h) lower surface mean-velocity profiles.

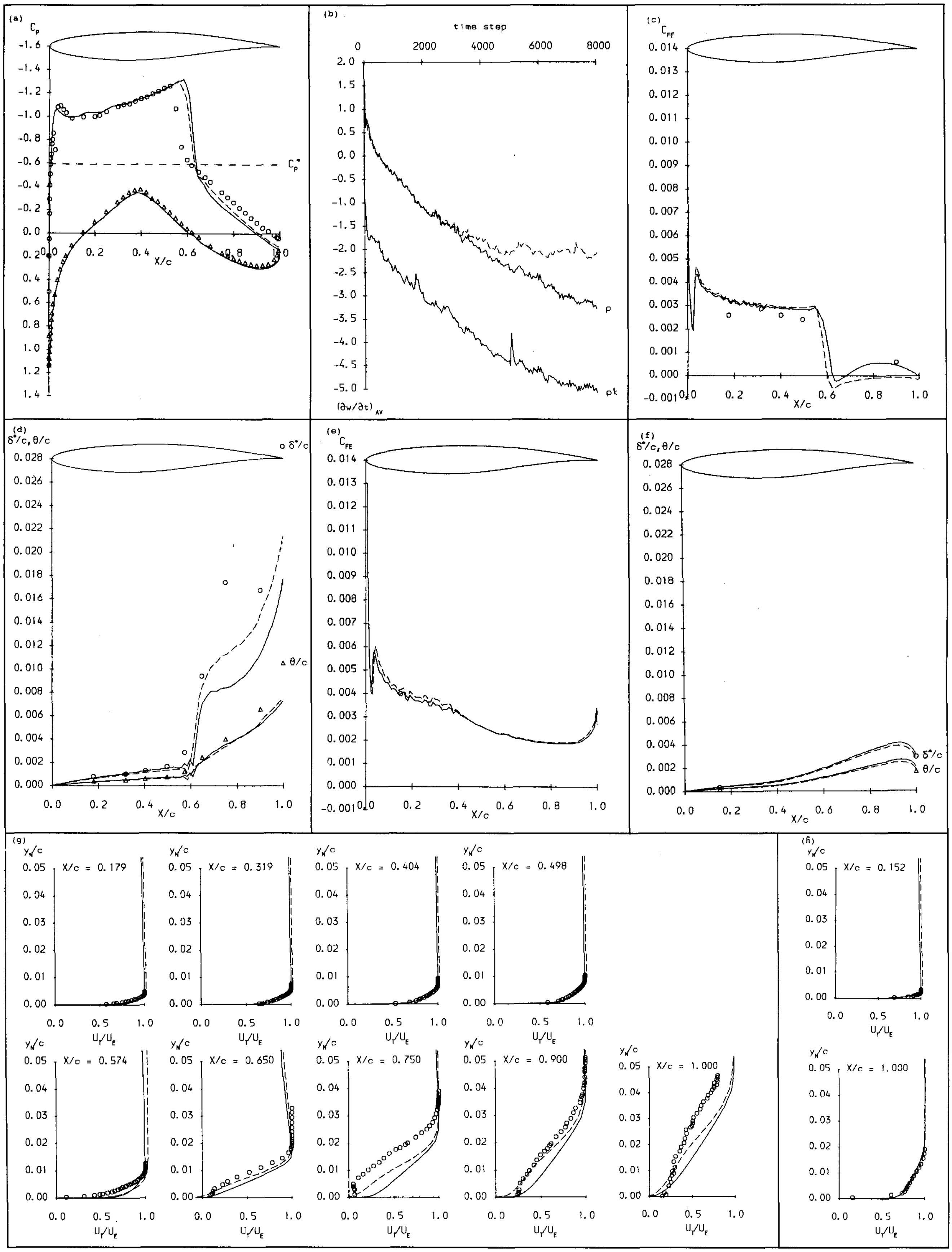

Figure 8. Results for RAE 2822 — case 10 — M = 0·75, α = 2·81° O, Δ experiment; ———— k–L_ϵ; — — — — Baldwin–Lomax: (a) surface pressure coefficient; (b) convergence history; (c) upper surface skin friction; (d) upper surface integral thicknesses; (e) lower surface skin friction; (f) lower surface integral thicknesses; (g) upper surface mean-velocity profiles; (h) lower surface mean-velocity profiles.

TABLE 2
RAE 5225 Aerofoil
Results using Baldwin–Lomax model
(B–L[1] run to α; B–L[2] run to C_L)

(a) Flow conditions

Case	M	α_{corr}	$R \times 10^{-6}$
A	0·598	1·81°	20
B1	0·735	1·59°	6·03
B2	0·737	2·76°	6·04

(b) Comparison of measured and predicted loads

Case A	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·433	—	—	0·00755	—
B–L[1]	0·4364	0·003600	0·003115	0·006715	−0·09215

Case B1	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·403	—	—	0·01068	—
B–L[1]	0·4792	0·005526	0·006158	0·01168	−0·1040
B–L[2]	0·4030	0·004999	0·006180	0·01118	−0·1038

Case B2	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·659	—	—	0·01292	—
B–L[1]	0·7184	0·01111	0·005845	0·01696	−0·1024
B–L[2]	0·6589	0·008239	0·005996	0·01424	−0·1017

Figure 4(b) shows the RAE 5225 aerofoil with a detail of the computational grid. The design geometry is used, since Ashill *et al*[10] indicate that only minor differences result from calculations using the manufactured geometry. Conditions for the three cases considered here are given in Table 2(a), transition being fixed at 5% chord on the upper and lower surfaces for all cases. Note that the incidence angle is measured relative to the axis about which the aerofoil geometry is defined, rather than the chordline joining the leading and trailing edges. The Baldwin–Lomax turbulence model is used for the calculations, since the flow appears to be fully-attached in all cases. Only experimental surface pressure distributions and integrated loads are available for comparison with calculations.

Figure 9(a) shows the results for Case A, a completely subcritical flow, at the higher Reynolds number of 20 million. The surface pressure distribution is very well-predicted, the only slight discrepancies being over the rear part of the lower surface. Consequently, the predicted lift and drag coefficients are in good agreement with experiment, Table 2(b).

Case B1, at the lower Reynolds number of 6 million, is a rather sensitive case, Fig. 9(b), with the upper surface at near-sonic conditions over the first 60% of chord. Overall, the surface pressure distribution is reasonably well-predicted using the Baldwin–Lomax model. The upper surface suction levels are a little too high, however, and the near-sonic plateau is terminated by a weak shock wave. This results in an over-prediction of the lift and drag coefficients, Table 2(b). A second calculation for Case B1, matching the experimental lift coefficient, results in a reduced incidence angle of 1·20° and some improvement in predicted drag coefficient. Figure 9(b) shows that the upper surface shock wave is now absent, but agreement with experiment deteriorates on the lower surface and in the leading-edge region.

The results for Case B2, at a similar Mach number but higher incidence angle than the previous case, are shown in Fig. 9(c). The surface pressure distribution is quite well predicted, particularly for the lower surface and the upper surface downstream of the shock wave. Again, the upstream upper surface suction levels are slightly overpredicted, and the terminating shock wave is too strong and positioned downstream of experiment. Matching the experimental lift coefficient, at a reduced incidence angle of 2·45°, leads to improvements in shock wave position and predicted drag coefficient, Table 2(b), with a small deterioration in the results for the lower surface.

The good predictions of lower surface pressures for Cases B1 and B2, at the corrected incidence angles given in Table 2(a), imply that the discrepancies on the upper surface are due to other causes. Further investigation is required to clarify this point.

8.3 CAST 7 Aerofoil

Stanewsky *et al*[30] present surface pressure measurements for the CAST 7 aerofoil over a range of flow conditions, including incidence sweeps at constant Mach number and a Mach number sweep at constant incidence. Test results from

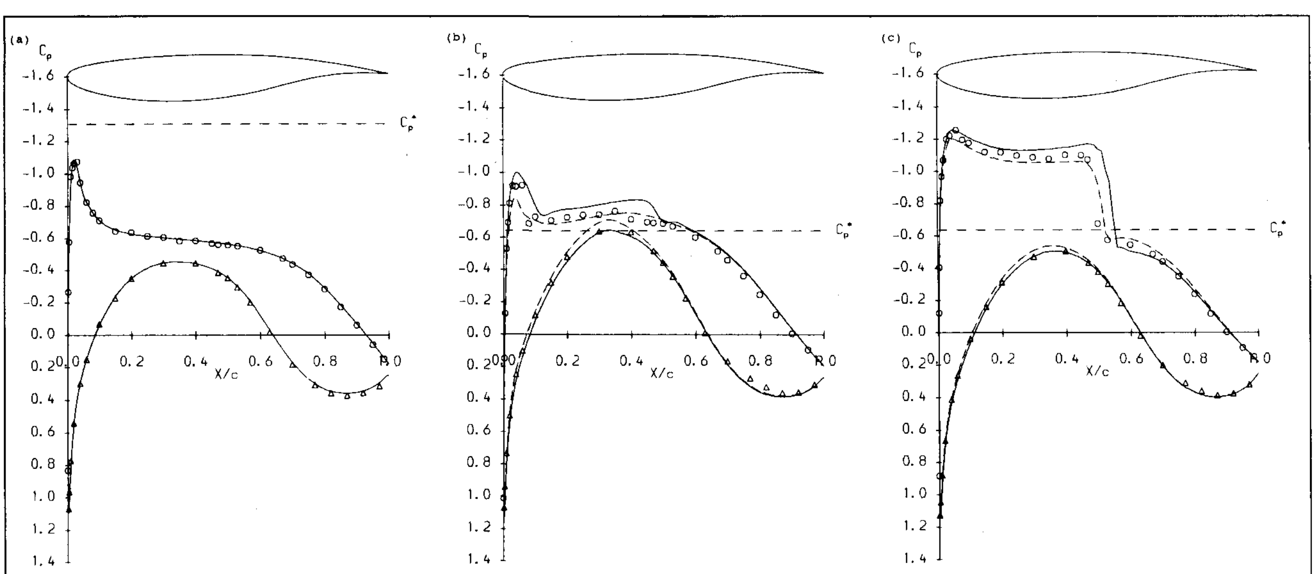

Figure 9. Results for RAE 5225 — Baldwin–Lomax model, O, Δ experiment; ——— run to α; — — — — run to C_L; (a) case A — M = 0·598, α = 1·81°; (b) case B1 — M = 0·735, α = 1·59°; (c) case B2 — M = 0·737, α = 2·76°.

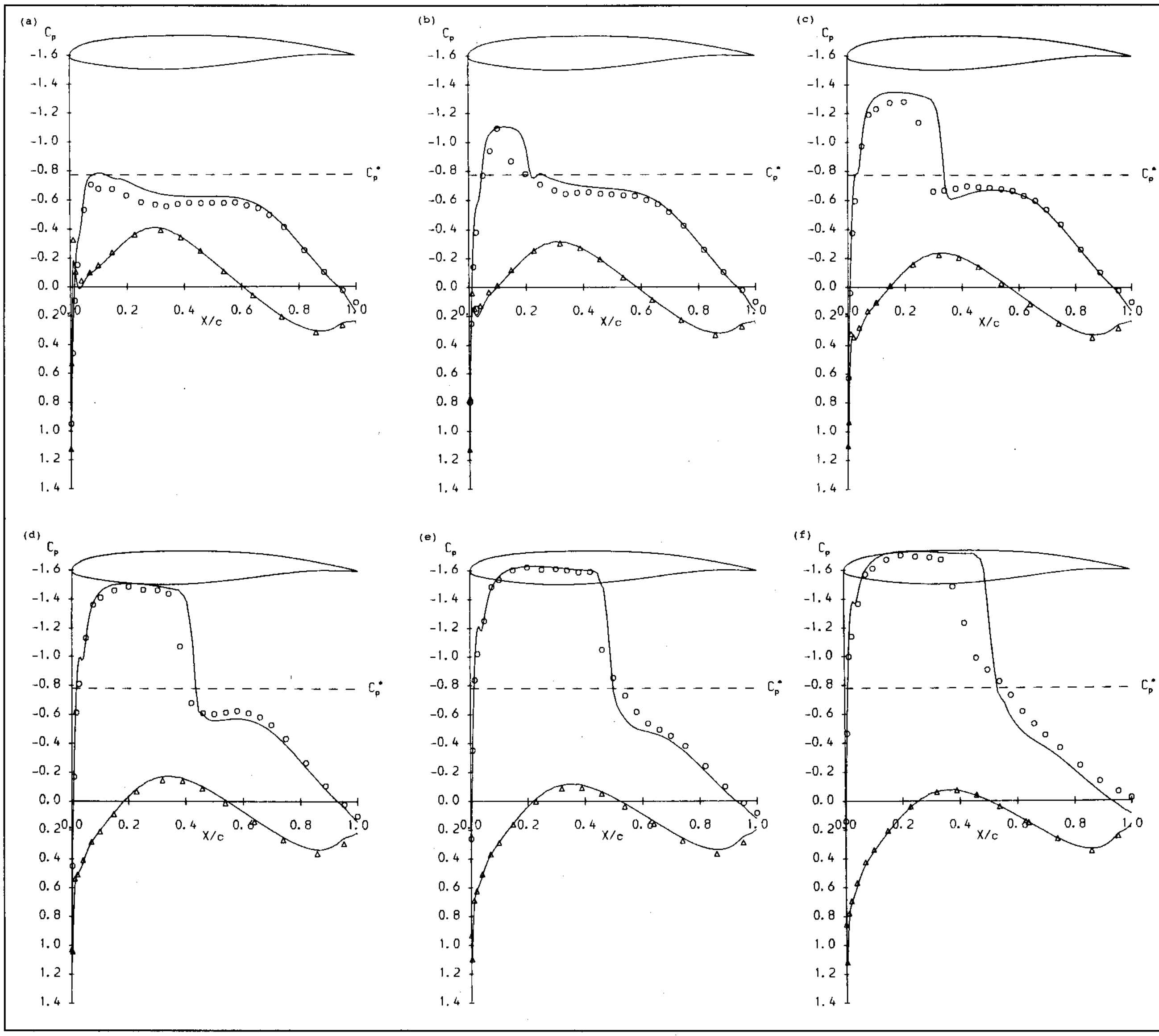

Figure 10. Results for CAST 7 — incidence sweep at M ≃ 0·70 O, Δ experiment; ——— k–L_ϵ;
(a) ARA run 58 — α = −0·19° (b) ARA run 59 — α = 0·76° (c) ARA run 60 — α = 1·71°; (d) ARA run 61 — α = 2·64°;
(e) ARA run 62 — α = 3·60°; (f) ARA run 63 — α = 4·59°.

the ARA Bedford 8 in × 18 in transonic wind tunnel are chosen for comparison, since the tunnel corrections to incidence angle, lift and pitching moment coefficients are well-defined for these cases[30]. The CAST 7 aerofoil has a maximum thickness of 11·8%, with a base thickness of 0·5%. The ARA model has an aspect ratio of 1·6, with a tunnel height/aerofoil chord ratio of 3·6.

Figure 4(c) shows the CAST 7 aerofoil geometry, together with a close-up view of the computational grid. The, presumably, design geometry given by Rizzi[27] is used, which incorporates a rather crude trailing-edge closure. Results are presented for the ARA incidence sweep at a Mach number of 0·70 and Reynolds number of 6 million, Table 3(a) listing the conditions for the individual cases. Transition is fixed at 8% chord on the upper and lower surfaces, in accordance with experiment. The k–L_ϵ one-equation turbulence model is used, since the RAE 2822 computations in Section 8.1 above indicate only minor differences from the Baldwin–Lomax model results.

Comparisons of measured and predicted surface pressure distributions are shown in Fig. 10, covering the range from essentially fully subcritical flow through to shock-induced separation. The general level of agreement between computation and experiment, particularly at high-lift conditions and on the lower surface, suggests that the procedures used to correct the incidence angle [30] are adequate. There are detailed discrepancies in upper surface pressure levels, however, with a consistent downstream prediction of the shock wave position. These, in turn, lead to an overprediction of lift and drag coefficients at a given incidence angle, Fig. 11(a) and Table 3(b). Nevertheless, the predicted lift-drag and lift-pitching moment polars, Fig. 11(b) and (c), are in good agreement with experiment.

Shock-induced separation is predicted for the two highest incidence cases, Fig. 10(e) and (f), over a region of $0{\cdot}48 < X/c < 0{\cdot}52$ and $0{\cdot}51 < X/c < 0{\cdot}59$ respectively. The k–L_ϵ turbulence model gives a very reasonable result for the former case, but with an overprediction of the shock wave strength. The large differences in shock wave position and the subsequent downstream pressure levels for the more extreme case of Fig. 10(f) indicate deficiences in modelling of the shock wave/boundary layer interaction.

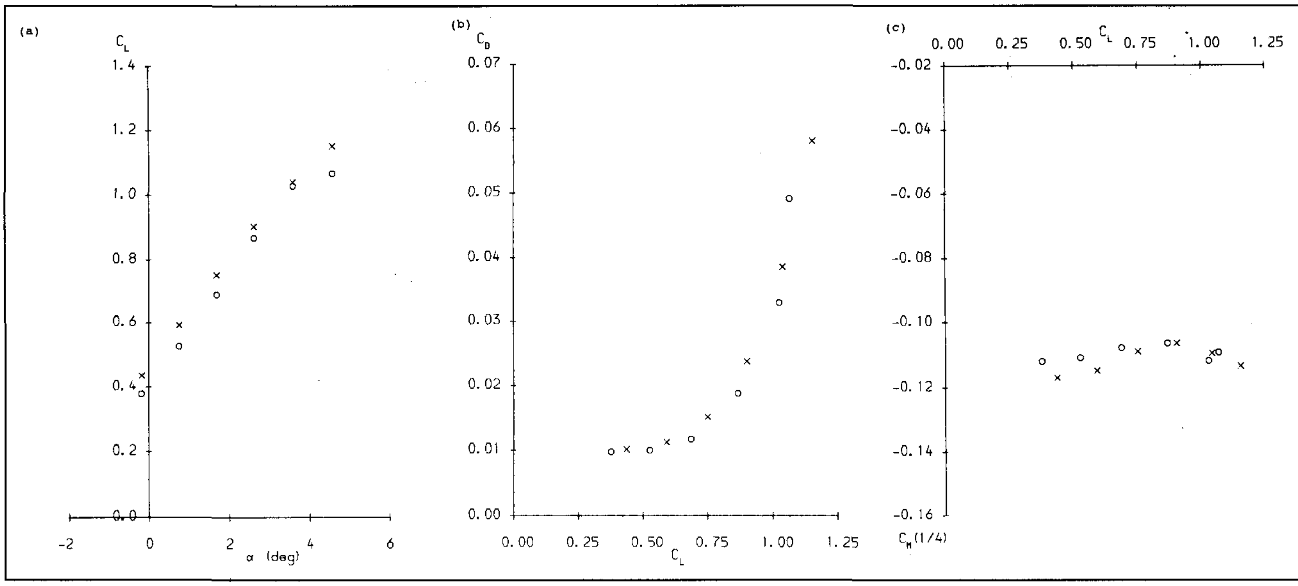

Figure 11. Results for CAST 7 — incidence sweep at M ≃ 0·70 O, experiment; x, k–L_ϵ;
(a) lift versus incidence; (b) drag versus lift; (c) pitching moment versus lift.

TABLE 3
CAST 7 Aerofoil — incidence sweep at M ≏ 0·70
Results using one-equation (k–L_ϵ) model

(a) Flow conditions

ARA run	M	α_{corr}	R × 10^{-6}
58	0·702	−0·19°	6·00
59	0·701	0·76°	5·98
60	0·701	1·71°	5·99
61	0·700	2·64°	5·97
62	0·700	3·60°	5·97
63	0·700	4·59°	5·96

(b) Comparison of measured and predicted loads

ARA run 58	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·379	—	—	0·00976	−0·1119
k–L_ϵ	0·4366	0·003871	0·006283	0·01015	−0·1169

ARA run 59	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·528	—	—	0·01001	−0·1108
k–L_ϵ	0·5928	0·004974	0·006261	0·01124	−0·1147

ARA run 60	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·688	—	—	0·01172	−0·1078
k–L_ϵ	0·7513	0·009046	0·006090	0·01514	−0·1089

ARA run 61	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·867	—	—	0·01879	−0·1064
k–L_ϵ	0·9025	0·01797	0·005851	0·02382	−0·1064

ARA run 62	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	1·027	—	—	0·03286	−0·1118
k–L_ϵ	1·0398	0·03295	0·005592	0·03854	−0·1095

ARA run 63	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	1·066	—	—	0·04912	−0·1092
k–L_ϵ	1·1519	0·05279	0·005310	0·05810	−0·1134

TABLE 4
MBB-A3 Aerofoil — incidence sweep at M ≏ 0·70
Results using one-equation (k–L_ϵ) model

(a) Flow conditions

ARA run	M	α_{corr}	R × 10^{-6}
84	0·700	1·82°	6·08
86	0·700	2·79°	6·26
88	0·699	3·72°	6·21
93	0·698	4·66°	6·17
94	0·698	5·15°	6·17

(b) Comparison of measured and predicted loads

ARA run 84	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·488	—	—	0·00785	−0·0461
k–L_ϵ	0·5459	0·004001	0·005500	0·009501	−0·04890

ARA run 86	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·632	—	—	0·00877	−0·0413
k–L_ϵ	0·7083	0·007418	0·005304	0·01272	−0·04172

ARA run 88	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·819	—	—	0·01561	−0·0338
k–L_ϵ	0·8624	0·01691	0·005064	0·02197	−0·03605

ARA run 93	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·977	—	—	0·03085	−0·0349
k–L_ϵ	1·0000	0·03349	0·004768	0·03826	−0·03719

ARA run 94	C_L	$C_D(P)$	$C_D(V)$	$C_D(T)$	$C_m(¼)$
experiment	0·997	—	—	0·03720	−0·0338
k–L_ϵ	1·0609	0·04432	0·004598	0·04891	−0·03950

The discrepancies in computed upper surface pressure levels and shock wave position are tentatively attributed to the very low aspect ratio 1·6 of the ARA wind tunnel model. As discussed by Stanewsky *et al*[31], the tunnel sidewalls can have a significant effect on the upper surface flow development for aspect ratios less than 2. Interaction of the aerofoil flow with the sidewall boundary layers, at low aspect ratios,

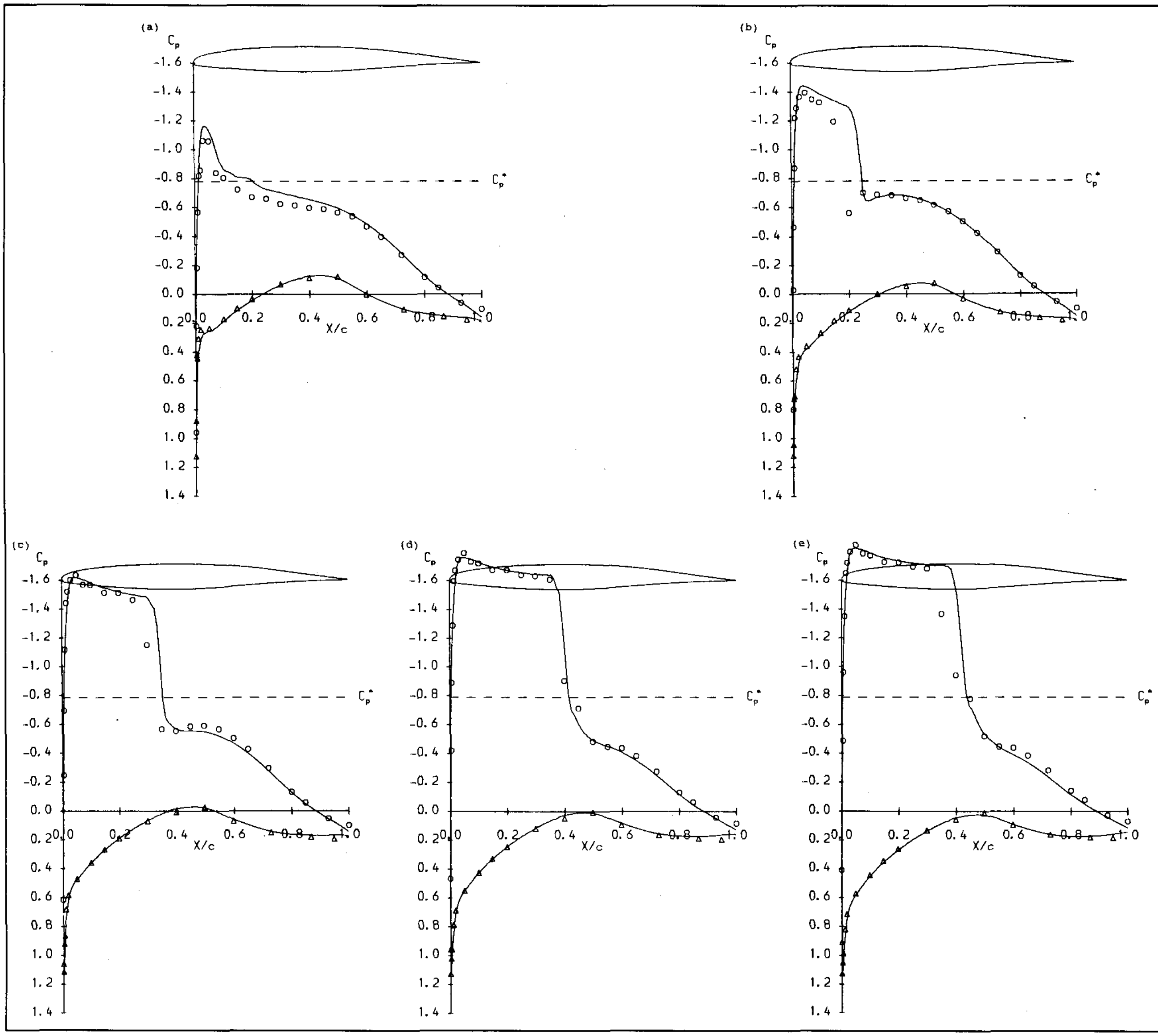

Figure 12. Results for MBB-A3 — incidence sweep at M ≃ 0·70 O, Δ experiment; ——— k–L_ϵ; (a) ARA run 84 — α = 1·82°; (b) ARA run 86 — α = 2·79°; (c) ARA run 88 — α = 3·72°; (d) ARA run 93 — α = 4·66°; (e) ARA run 94 — α = 5·15°.

tends to move the shock wave upstream compared to results for higher aspect ratio models. This is consistent with the present downstream prediction of the shock wave position compared to experiment. It would be of interest to compute the three-dimensional flowfield for the CAST 7 aerofoil in the ARA wind tunnel, including the sidewalls. Such calculations have been performed by Swanson *et al*[32] for the related CAST 10 aerofoil section, and these appear to support the experimental observations of Stanewsky *et al*[31].

8.4 MBB-A3 Aerofoil

As a final example, the MBB-A3 aerofoil is considered since, with a maximum thickness of 8·9% and a sharp trailing edge, it is the thinnest supercritical section for which experimental data is readily available in the literature. Bucciantini *et al*[33] report surface pressure measurements and integrated loads for a range of flow conditions in two different wind tunnels. Results are presented here for an incidence sweep at a Mach number of 0·70 and Reynolds number of 6 million, in the ARA Bedford 8in × 18in transonic wind tunnel. The aerofoil model has an aspect ratio of 1·6 and a tunnel height/aerofoil chord ratio of 3·6.

The ARA tests involve free transition, but it is noted[33] that transition is likely to occur near the leading edge, due to disturbances created by the surface pressure tappings. In any event, transition occurs upstream of the shock wave on the upper surface, since the observed shock wave/boundary layer interaction is turbulent in nature. For this reason, the present calculations assume fixed transition at 3% chord on the upper surface. The initially favourable pressure gradient on the lower surface suggests that a more aft transition position may be appropriate, and so transition is fixed at 40% chord on the lower surface. The aerofoil design geometry is used, and is shown in Fig. 4(d) together with a close-up of the computational grid. Table 4(a) lists the flow conditions for the various cases in the incidence sweep, and Table 4(b) compares measured integrated loads with predictions using the k–L_ϵ one-equation turbulence model. The measured incidence angles, lift and pitching moment coefficients are in corrected form, following the procedures recommended for the ARA tunnel[33].

Predicted surface pressure distributions for the five cases are compared with experiment in Figure 12. The level of agreement is similar to that achieved for the CAST 7 aerofoil

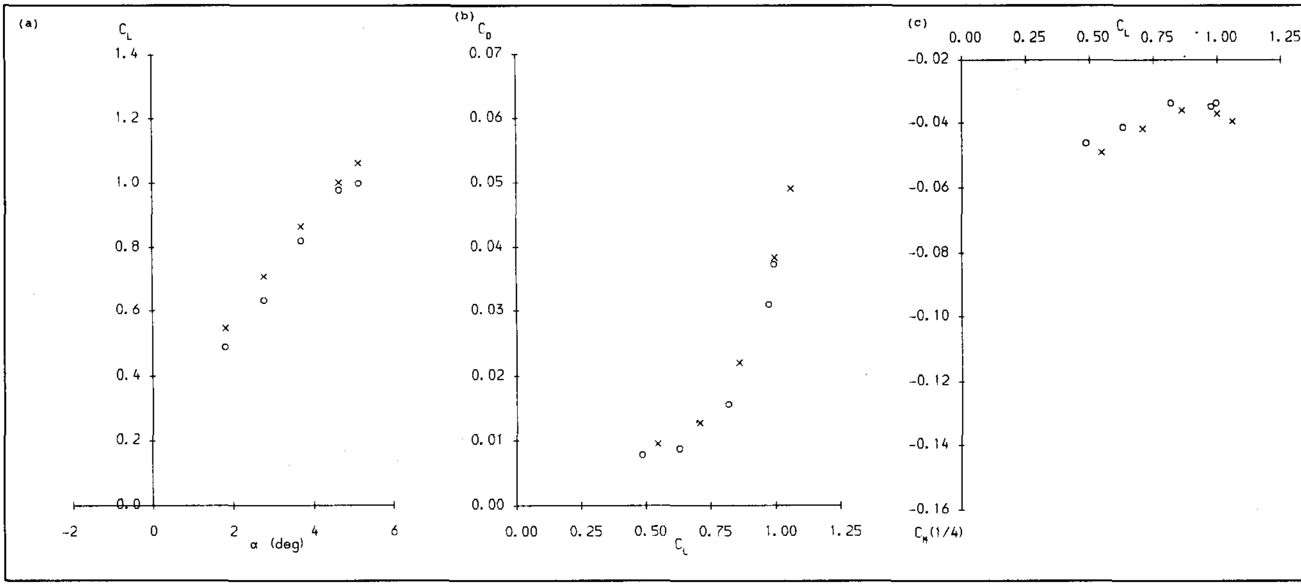

Figure 13. Results for MBB-A3 — incidence sweep at M ≃ 0·70 O, experiment; x, k–L_ϵ; (a) lift versus incidence; (b) drag versus lift; (c) pitching moment versus lift.

cases considered above, involving tests in the same ARA wind tunnel. Differences at the higher incidence angles are confined to the shock wave position, which is predicted downstream of experiment. The discussion in Section 8.3 above concerning model aspect ratio effects appears also to be relevant to these MBB-A3 tests. Shock-induced separation is predicted for the two highest incidence cases, Fig. 12(d) and (e), with chordwise extents of $0{\cdot}4 < X/c < 0{\cdot}45$ and $0{\cdot}42 < X/c < 0{\cdot}5$ respectively. Fully-attached flow at the trailing edge is predicted for all cases.

The downstream position of the shock wave compared to experiment leads to an overprediction of lift coefficient for a given incidence angle, Fig. 13(a). Lift-drag and lift-pitching moment polars, Fig. 13(b) and (c), are reasonably well predicted, but agreement with experiment for the former is not as good as that achieved for the CAST 7 aerofoil.

9. CONCLUSIONS

A method to predict the transonic viscous flow development around aerofoil sections has been developed, based on a solution of the Reynolds-averaged Navier–Stokes equations. The finite volume spatial discretisation and time-marching procedures adopted for the mean-flow equations are readily extended to include a modelled transport equation, for the turbulent kinetic energy. It is found that no changes are necessary to the basic flow solver when this additional equation is incorporated, and convergence characteristics are unaffected. Turbulence modelling is currently at the isotropic eddy-viscosity level, with a choice of either the Baldwin-Lomax algebraic model or a k–L_ϵ one-equation model.

Detailed comparisons of predictions with experiment for the RAE 2822 aerofoil indicate minor differences in results using the two turbulence models. However, the tendency of the Baldwin–Lomax model to predict premature flow separation, particularly at the foot of a shock wave, is noted. The k–L_ϵ is well-behaved in such circumstances, and also gives reasonable predictions in the presence of moderate shock-induced separation. Agreement with experiment deteriorates for more extensive separated flow, and this is attributed to limitations in the current level of turbulence modelling with the one-equation model. However, scope for significant improvement does exist, by the incorporation of non-isotropic modelling for Reynolds stresses via an algebraic Reynolds stress closure. It is not obvious that such turbulence-modelling refinements can be implemented in the algebraic Baldwin–Lomax model. For this reason, the use of the one-equation model is recommended for transonic aerofoil flows, and turbulence model improvements should be pursued.

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